□□□□ □□ □□□ v12 (Integrated Experiments)

□□ □□ □□□ □□ □□□ □□ □□□ □□□ □□ Triple-Hybrid RAG □□□□□□ □ □□ □□□ □□

Performance Optimization of Triple-Hybrid RAG Framework via Proximal Policy Optimization-based Learned Dynamic Weighting

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□□ □□ □□(RAG)□ □□ □□ □□(LLM)□ □□□ □□ □□ □□□ □□□□ □□ □□□□, □□ □□ □□ □□□ □□□ □ □□□ □□ Vector, Graph, Ontology □ □□□ □□□ Triple-Hybrid RAG □□□□□□ □□□□□. □ □□ □ □□ □□ Shin & Moon (2025, JKSCI) □ Rule-based Dynamic Weighting Algorithm (R-DWA) □□ □ □□□ □□□ (α, β, γ) □ 2 □□ □□□□ □□□□□□, □□ □□□ □□□ □□ (□□□ □□, □□ λ, □□□ □□ □□ □□ □□, □□□ □□ □□) □ □□□.

□ □□ □□□ 3 □□ □□ □ □ □□□ □□□□.

(1) □□□ □□. □□ JKSCI □□□ □□□ □□ 3 □□ — (i) evaluator.py □ □□□ □□ □□□□ □□ F1 □□ □□, (ii) Table 8 □□ □□□□ □□□ □□ (execution_count: null), (iii) gold □□□ □□□ LLM □□ □ □□□ □□ □□ — □ □ □□□ □□□□□ □□·□□□□ □□ □□□□ CORRIGENDUM □ □□□□□. □□□ □□ □ □□ □ □□□ (corrected baseline: S1+S2+S3 □□ □□□ R-DWA, F1_{strict} 0.529) □ □□□□□.

(2) □□□ □□. Triple-Hybrid RAG □ □□□ □□□ 18 □□ □□ × 3-simplex □□□ 1-step Markov Decision Process □ □□□□□, Proximal Policy Optimization (PPO) □□ Actor-Critic □□□□ (5,636 □□□□) □ □□□□ Learned Dynamic Weighting Algorithm (L-DWA) □ □□□□. □□ □□ □□ □□□ □□ 330,000 □□□ □□□□ □□ □□ □ □□□□ PPO □□ □ LLM API □□□ 0 □□ □□□□ 3 seeds □□ □□□ $288 □□ $33 □□ 85% □□ □□□.

(3) □□ □□. Corrected baseline □□□ 5,000 QA × 3 seeds □□ □□, L-DWA □ R-DWA □ □ F1_{strict} 0.529→0.562 ± 0.007 (+6.2%), F1_{substring} 0.482→0.507 ± 0.009 (+5.2%), Faithfulness 0.544→0.580 ± 0.009 (+6.6%) □ □□□ □□□ □□□□□. □□ L-DWA □ Dirichlet □□□ per-query argmax Oracle (66-□ □□ □□ □□) □ □□ F1 □□□□ □□ □□ (0.562 > 0.554) — □□ □ □□ □□□ □□ argmax □ □□□ □□□□ □□ □□ □□□ □ □□□□. □□□ □□ □□□□□ F1_{strict} +36.7% (0.223→0.304) □ Oracle □ □□□□ per-query □□□ □ □□□ □□ □□□□ □□□□.

□□ □□□ □□. HotpotQA/MuSiQue/PubMedQA □□□ naive transfer □□ (−30~35%) □ □□□ L1 □□ □□·L2 □□□ □□·L3 Graph/Ontology □□□ 3-□ □□ □ □□□□□. □□ intent □□ □□ (L1 □□) □ R-DWA +38%, L-DWA +49% □□□ □□□□□□, □□ □□ □□ □□□□ (403 QA) □□ □ □□□ R-DWA □ F1_{strict} 0.663 (Korean 0.529 □□ +25%) □ □□□□ L-DWA □□□□□ □□ □□□ □□ □□□□□. □ □□□ □□□ cross-domain □□□ □□□□ □□ □□ □□□ alignment □□ □ □□□□□□□.

□□ □□·330K □□·3-seed □□□□□·Streamlit □□□□□ GitHub (sdw1621/triple-rag-phd) □ □□□□ □□ □□□□ □□ □□ □□□□.

□□□: Triple-Hybrid RAG, □□□□, PPO, □□ □□□, □□□□ □□, □□□ QA, □□ □□□, □□□□, □□□, □□□□□□

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Abstract

Retrieval-Augmented Generation (RAG) is a key technique for mitigating hallucination and knowledge-cutoff issues in Large Language Models (LLMs), and a Triple-Hybrid RAG architecture combining Vector, Graph, and Ontology sources was recently proposed to overcome the limits of single-source retrieval. The author's prior work, Shin & Moon (2025, JKSCI), introduced the Rule-based Dynamic Weighting Algorithm (R-DWA) that determines the three source weights (α, β, γ) via a two-stage rule, but the rule-based nature imposes structural limits (domain bias, single-parameter λ, lack of conditional-query resolution, no cross-domain transfer).

This dissertation addresses these limits via three-layer contributions.

(1) Reproducibility. The author identified and voluntarily corrected three reproducibility defects in the prior JKSCI paper: (i) a punctuation-stripping bug in evaluator.py that shrank list-form F1 to roughly one-third of its true value; (ii) the absence of execution evidence (execution_count: null) for the Table 8 notebook cell that claimed 5,000 QA × 3 runs; and (iii) misalignment between list-form gold answers and sentence-form LLM outputs. A CORRIGENDUM disclosing all three issues has been published at the prior repository. These corrections establish a trustworthy new baseline (corrected R-DWA, F1_{strict} = 0.529) against which the core contributions of this work are measured.

(2) Methodological. This work formalizes the weight-decision problem as a one-step Markov Decision Process (18-dimensional state × 3-simplex action) and proposes Learned Dynamic Weighting Algorithm (L-DWA), which trains an

Actor-Critic network (5,636 parameters) via Proximal Policy Optimization (PPO). A 330,000-entry offline reward cache pre-computes all (query, discretized-weight) rewards, reducing a 3-seed training budget from \$288 to \$33 (−85%) while keeping LLM API calls during PPO training at zero.

(3) Empirical. On the corrected baseline, L-DWA outperforms R-DWA by F1_{strict} +6.2% (0.529→0.562 ± 0.007, 3-seed std < 1.2%), F1_{substring} +5.2% (0.482→0.507), and Faithfulness +6.6% (0.544→0.580). Most importantly, L-DWA's continuous Dirichlet mean exceeds the per-query argmax Oracle (defined over a 66-point discrete grid) on every F1 metric (F1_{strict}: 0.562 > 0.554; F1_{sub}: 0.507 > 0.504; F1_{char}: 0.494 > 0.487), demonstrating that the learned continuous policy escapes the theoretical ceiling of discrete argmax. The conditional-query subtype shows a qualitative gap of +36.7% over R-DWA (0.223→0.304), likewise exceeding Oracle (0.290).

Cross-domain analysis. Naive transfer to HotpotQA / MuSiQue / PubMedQA shows a −30~35% degradation of the university-trained L-DWA. This work decomposes the failure into three causal layers: L1 language barrier, L2 domain vocabulary, and L3 absence of Graph/Ontology. Adding English intent patterns (addressing L1) recovers +38% for R-DWA and +49% for L-DWA. A newly constructed English synthetic university benchmark (403 QA) shows R-DWA reaching F1_{strict} = 0.663, +25% above the Korean equivalent, demonstrating that the L-DWA architecture is language-neutral and that cross-domain failure is a data-alignment issue, not an architectural limit.

All code, the 330K-entry reward cache, 3-seed checkpoints, and a Streamlit dashboard are publicly available at github.com/sdw1621/triple-rag-phd for one-command reproduction.

Keywords: Triple-Hybrid RAG, Reinforcement Learning, Proximal Policy Optimization, Learned Dynamic Weighting, Offline Reward Cache, Korean QA, Knowledge Graph, Ontology, Reproducibility, Cross-lingual Transfer

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Abstract

□I□ □□

- 1 □ □□ □□
- 2 □ □□ □□ (RQ1~RQ3)
- 3 □ □□ □□ (□□□ 2 + □□□□□ 2 + □□ 4)
- 4 □ □ □□□ □□□ □□
- 5 □ □□ □□
- 6 □ □□□ □□

□II□ □□ □□

- 1 □ RAG □ □□□ □□□□□ □□
- 2 □ □□ □□□ □ □□□□□ RAG □□
- 3 □ □□·□□□ □□□ □□□□ □□ □□□
- 4 □ □□□ QA □□ □ JKSCI 2025 □□□
- 5 □ □ □□□ □□□ □□

□III□ Triple-Hybrid RAG □□□□

- 1 □ □□ □□□□□
- 2 □ □ □□ □□ (Vector α, Graph β, Ontology γ)
- 3 □ QueryAnalyzer

- 4 □ Score Fusion: merge_contexts
- 5 □ Prompt Template
- 6 □ Pluggable DWA Interface

□IV□ □□ □□ □□ □□□ □□□□ (R-DWA)

- 1 □ R-DWA □□ □□ (2 □□)
- 2 □ R-DWA □ □□
- 3 □ R-DWA □ □□□ □□ (4 □□)
- 4 □ R-DWA □□ □ □□
- 5 □ R-DWA □ □ □□ □□□ □□
- 6 □ □□

□V□ □□□ □□ □□□ □□□□ (L-DWA)

- 1 □ MDP □□□
- 2 □ Actor-Critic □□□□ (5,636 □□□□)
- 3 □ PPO □□ □□□□
- 4 □ □□□□ □□ □□ ($33, 330K □□□)
- 5 □ □□ □□□ □□
- 6 □ □□

□VI□ □□ □ □□

- 1 □ □□ □□ □ □□ □□ (□□ F1 □□)
- 2 □ □□□□ □□ □□ □□ □□
- 3 □ □□ □□ □□ (5,000 QA, 3 seeds)
- 4 □ □□ □□□ □□ □□ (Conditional +131%)
- 5 □ PPO □□ □□□
- 6 □ □□ □□
- 7 □ □□ □□□ □□ (HotpotQA / MuSiQue / PubMedQA)
- 8 □ □□ □ □□ □□
- 9 □ □□

□VII□ □□

- 1 □ □□ □□

- 2 □ □ □□□ □□
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□□ **A.** □□ □□

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□I□ □□ (v5 □□□)

1 □ □□ □□

□□ □□ □□(Retrieval-Augmented Generation, □□ RAG)□ □□ □□ □□(LLM)□ □□□ □□ □□ □□□ □□ □□ □□□□ □□□□ □□ □□□□, 2020 □ □□ □□·□□ □□□□ □□□ □□□□□. □□ □□ RAG □ □□□ □□□□□ □□□ □□ □□(□□□ □□, □□ □□□, □□ □□□□)□ □□□□ □□□□□ RAG □ □□□□□□, □ □□□ □□ □ □□ □□(Shin & Moon, 2025, JKSCI)□□ Triple-Hybrid RAG □□□□□□ Rule-based Dynamic Weighting Algorithm (R-DWA) □ □□□ □ □□.

□□ □□□ R-DWA □ □□ □□ □□□ □□/□□/□□ □□ □□□ 2 □□□ □□□□ □ □□□ □□□ (α, β, γ)□ □□□□ □□□□. □□ □□ □□ □□□□(5,000 QA) □□ R-DWA □ Vector □□ □□ □□ F1 □ □□□□□ □□□□□□, □□ □□□□□ □□□ □□ □□□ (i) □□ □□□ □□□□ □□□ □□□□ □□□ (ii) □□ □□□ □□□(Table 4-1)□ □ □□ □□□ □□□□ (iii) □□□ □□(conditional query)□□ □□ □□□ □□ □□ □□□□ □□□□.

□ □□ □□□ □ □□□ □□□ □□□ □□ □□□ □□□□(Learned DWA, □□ L-DWA) □ □□□□. □□□ □□□ 1-step contextual bandit □□□ Markov Decision Process □ □□□□□, Proximal Policy Optimization(PPO)□□ □□□ □□□□. □□□ □□ □□□ □□□□ □□ □□□ □□□□, □□ □ LLM API □□□ 0 □ □□ □□□ □□□□.

2 □ □□ □□

□ □□□ □□ □ □□ □□ □□(RQ)□ □□□.

RQ1. □□ □□ R-DWA □ □□□ □□ □□□ □□□□ □□ □□□□□ □□ □□(reward landscape)□ □□□□ □□□? □□□□ □□□□ □ □□□ □□□□□ □□□□?

RQ2. Triple-Hybrid □□□ □□□ MDP □ □□□□□ PPO □ □□□ L-DWA □ R-DWA □□ □□ □□□ □ □□ □□□□□? □□□ □□ □□□□ □□□ □□□□□?

RQ3. □□ □□□(□□ □□)□□ □□□ L-DWA □ □□□ □ □□□(HotpotQA, MuSiQue, PubMedQA) □□ □□ □□□□? □□ □□□/□□□□□ □ □□□□□□ □□ □□□ □□ □□□□ □□□?

3 □ □□ □□

□ □□□ □□□ □□ □ □□□ □□□□.

3.1 □□□□ □□ (Methodological)

1. MDP □□□. Triple-Hybrid RAG □ □□□ □□□ 18 □□ □□ □□ × 3-simplex □□ □□□ 1-step MDP □ □□□□ □□□ □□□ (Ch.5 §1).
2. PPO-based L-DWA. Actor-Critic □□□□ (shared backbone 18→64→64, Dirichlet □□ □□, 5,636 □□□□) + Clip 0.2 PPO □□ □□□□ 3 seeds (42, 123, 999) □□ □□ □□ 0.215 ± 0.002 □ □□□□ □□ □□□□□ □□□□ □□□□ (Ch.5 §3, Ch.6 §5).

3.2 □□□□□ □□ (Engineering)

3. □□□□ □□ □□. 5,000 □□ × 66 □□ □□□ = 330,000 □□□□ reward □ SQLite □ □□ □□ ($33, 14 □□)□□ PPO □□ □ API □□ 0, 3 seeds □□□ 50 □□ → 1 □□ (−98%) □□ □□□□ (Ch.5 §4).
4. □□ F1 □□ □□. □□□ □□·□□□ □□ □□ □□ F1 (F1_strict) □ □□ □□□ gold □ □□□□ □□□□□ F1 (F1_substring)□ □□□□, □□ □□□□ □□□□ □□□ □□□□ □□□ □□□□ (Ch.6 §1.3).

3.3 □□□ □□ (Empirical)

[□□ 5] □□□ □□ □□□□□□ L-DWA □□ □□ (5,000 QA, 3 seeds, sentence □□□□ □□):

- F1_strict 0.167 ± 0.001 (R-DWA 0.137 □□ +21.9%)
- F1_substring 0.488 ± 0.006 (+8.4%), Faithfulness 0.866 ± 0.004 (+3.6%)
- L-DWA □ per-query argmax Oracle □□ F1_strict 99.4% □□, F1_substring □□□ Oracle □□ (0.488 > 0.483).

[□□ 6] □□□□ □□ □□ (list prompt, 5,000 QA, 3 seeds): gold □□□ □□□ □□

PROMPT_TEMPLATE_LIST □□ □:

- R-DWA F1_strict 0.529, L-DWA 0.562 ± 0.007 (R-DWA □□ +6.2%)
- L-DWA □ Oracle □□ 0.554 □ □□ — □□ Dirichlet □□□ 66-□ □□ □□□ argmax □ □□□ □ □□.
- EM □□□ 0 → 0.388 □ □□ (Ch.6 §3.4).

[□□ 7] Conditional □□□□□ □□ □□ (n=1,250):

- R-DWA F1_substring 0.090 → L-DWA 0.208 (+131%), Oracle 0.195 □□ (Ch.6 §4).
- □ □□□ □□ □□□ per-type claim.

[□□ 8] □□ □□□ □□□ □□ □□ + □□ (Ch.6 §7): □□-□□ L-DWA □

HotpotQA/MuSiQue/PubMedQA naive transfer □□ −30~35% □□ □□. □ □□□ □ □□□

□□·□□:

- L1 (□□ □□): □□ intent □□ □□ □ R-DWA +38%, L-DWA +49% □□ (Option A).
- L2 (□□□ □□): □□ □□ □□ (403 QA) □□ □ R-DWA F1_strict 0.663 □□ — Korean 0.529 □□ +25%. □□□ □□□ □□ □□□ (Option B).
- L3 (Graph/Ontology □□): HotpotQA □□ L3 □ □□□□□□ □□ □□ — Triple-Hybrid □ □ □□ □□ □□.
- Option C: □□ □□ L-DWA □□ □□ Korean-only state □□□ L-DWA ≈ R-DWA — state □□□□ L-DWA □□□ □□ □□ □□.

□□ 8 □ □□□ □□: □□ "L-DWA naive transfer □□" □□□ "□□□ 3-□ □□□ □□ □□, □ □□□ □□ □ □□ □□□□□ □□" □ □□□□. □□ □□ (□□□ intent encoder, domain-invariant state) □ □□ □ □ □□.

4 □ □ □□□ □□□ □□

□ □□□ □□□ □□□:

- Triple-Hybrid RAG (Vector + Graph + Ontology) □ □□□ □□ □□
- PPO □□ □□ □□□ □□□ □□□
- □□ □□ □□□□□ 5,000 QA □□ □□

- HotpotQA/MuSiQue/PubMedQA □□ □□□ □□ □□ (domain transfer feasibility)
- □□-loose □□ □□ □□□ □□

□ □□□ □□□ □□□ □□□ (□□ □□ □□):

- □□ □□□·□□□□□ LLM □□ □□ □□
- □□□ intent □□□ (□ □□□ □□□ regex □□)
- □□□ □ joint training □□ meta-learning
- BERT multi-label intent classifier □ □□ (§5.2 □□ □□□ □□□□ intent_logits=0 □□□ □□)
- HotpotQA □□□ L-DWA domain-specific □□ (reward landscape cliff □ □□ □□ □□ □□□ Ch.6 §7.3 □ □□)

5 □ □□ □□

- □I□ (□ □) □□ □□, □□ □□, □□, □□
- □II□ □□ □□ — RAG / Graph-RAG / Hybrid-RAG / PPO / □□□ QA □□
- □III□ Triple-Hybrid RAG □□□□ — Vector·Graph·Ontology □ □□□ □□□ □□ □□
- □IV□ Rule-based DWA (R-DWA) — □□ □□□ 2 □□ □□ □ □□ □□
- □V□ Learned DWA (L-DWA) — MDP □□□, Actor-Critic, PPO □□, □□□□ □□
- □VI□ □□ □ □□ — □□ F1 □□, 3-seed □□, □□ □□□ □□, □□ □□□ □□
- □VII□ □□ — □□ □□, □□, □□ □□ □□

6 □ □□□ □□

□ □□□ □□ □□, □□□□ □□, PPO □□□□□, □□ □□□ GitHub □□□□□ □□□□:

- □□□: https://github.com/sdw1621/triple-rag-phd (Private → □□ □ Public)
- □□ □□ □□□□: scripts/ □□□□ (build_cache, train_ppo, evaluate_rerun, cross_benchmark □)
- □□: cache/university.sqlite (330K □□□, 16.8 MB)
- □□□□□: cache/ppo_checkpoints/seed_{42,123,999}/final.pt
- □□ □□ markdown: thesis_current/v5_rewrite/

□□ □□ □ □□□ □□ □□□□ □□□ □□□□, PyTorch deterministic □□ □ 3 □ □□ □□ (42, 123, 999) □ □□□ □□ □□□ □□□□.

□□ □□ (v4 → v5)

□□	**v4 (□□)**	**v5 (□□)**
□□ F1 □□	0.089 □□	F1_strict 0.167 / F1_substring 0.488 (□□, □□ □□ □)
□□ □□	+3.5% F1	+21.9% F1_strict / +8.4% F1_substring
Boundary query	Boundary EM +32.8%	Conditional F1_substring +131%
□□□ □□	"Zero-shot □□"	"Domain-specific learning □□□ □□"
Oracle □□	□□	Oracle 99.4% □□ □□
□□ □□	F1 □□	F1_strict + F1_substring □ □
□□	Boundary / Zero-shot	Conditional / Domain-specific

v4 □ □□□ □□□ □□□ □□ □□□□ □□□□, □□□□ □□ (MDP + PPO + Cache + □□ □□) □ □□□ □ □□□ □□□□ □□□□□□.

□II□ □□ □□ (v5 □□□)

□ □□ □ □□□ □ □□ □□ □□ □□□ □□□□: (1) RAG □□□ □□□□□ □□, (2) □□ □□□·□□□□□ RAG □ □, (3) □□/□□□ □□□ □□□□□□ □□□□□ □□. □□□ □□□ □ □□□ □□□ □□□□.

1 □ RAG □ □□□ □□□□□ □□

1.1 □□ □□ RAG □ □□

Lewis et al. (2020)□ RAG □□□□ Dense Vector Retrieval □ □□ □□□ □□□□, LLM □□ □□ □ □□□ □□□□. □□□ □□ □□ □□□ (i) □□□ □□ (multi-hop) □□ □□ □□□□ □□□□, (ii) □□ □□ (e.g. "40 □ □□") □ □□□□□ □□□□ □□□, (iii) □□ □□□□ □□□ □□□ □□□ □□□ □□□ □□□□□ □□□ □□□ □□□□□ □□□□□ (Self-RAG, Asai et al. 2023; CRAG, Yan et al. 2024).

1.2 GraphRAG □□

Microsoft GraphRAG (Edge et al. 2024) □ □□ □□□ □□ → □□□ □□□□ □□ → □ □□□□ □□ □ □□□□ □□□ multi-hop □□□□ traditional RAG □□ 34% □□□ □□□□□. LightRAG (2024), Nano-GraphRAG □ □□□ □□□ □□□□□, 2026 □ 4 □□□ □□□□□□ Graphify

(safishamsi/graphify, 2026) □ □□□□ □□□□ "□□□ □□ □□ □□□ LLM □□□□□ □□□ □□ □□" □ □□□□. □ □□□ □□ □□ (Shin & Moon, 2025) □ Graph □□□ □□·□□□□□ □□□ Triple-Hybrid □□□ □□□ □ □□.

1.3 Hybrid RAG

HybridRAG (2024) □ vector retrieval □ graph retrieval □ □□ □□□ □ □□□□ □□ □□□□□ □□□□□. □ □□□ □□ □□□ Triple-Hybrid RAG □ Vector + Graph + Ontology □ □□□ □□□ □□□ □□□□, □□ □□□ □□ □□□□ □□ □□ □□□□□ □□□□ □□□□□.

2 □ □□ □□□ □ □□□□□ RAG □□

2.1 □□ □□□ □□ RAG

NetworkX BFS □□ Neo4j Cypher □□□ □□ □□□ RAG □ □□□□ □□□ multi-hop QA □□□ □ □□ □□ □□□ □□ □□□□ □□□□. □ □□□ NetworkX MultiDiGraph □□ BFS(max_depth=3)□ Graph □□□ □□□□ (Ch.3 §2.2), 577 □ □□ / 60 □□ / 400 □□□□ □□□ □□ □□ □□□□ □□□□□.

2.2 □□ □□□□□ RAG □□

OWL 2 □□□ HermiT/Pellet □□□□ □□□ □□, □□ □□, □□ □□□ □□ □□□ □□□ □□□□. □ □□□ Owlready2 □□ □□□□□ Ontology □□□ □□□□ (Ch.3 §2.3), □□ □□□ □□ (conditional query) □ □□ □□ ("40 □ □□") □ □□ □□□ □□ □□□□.

3 □ □□·□□□ □□□ □□□□ □□ □□□

3.1 Adaptive-RAG □ □□ □□ □□

Jeong et al. (2024) Adaptive-RAG □ □□ □□□□ □□ □□ □□/□□ □□/no-retrieval □ □□□ □□ □ □□□□ □□□□. □□□ □□ □□ □□ □□ □ □□ □□□ □□ □ □□□ □□□.

3.2 Instruct-RAG / Self-RAG / CRAG

Self-RAG (Asai et al. 2023) □ CRAG (Yan et al. 2024) □ retrieve/generate □□□ self-reflection □□ □□ corrective passage selection □ □□□□. □□□ □□ □□□□ □□□□ quality control □ □□□ □□□, □□ □ □□□ □□ □ □□ □□□□ □□□□ □□□.

3.3 PPO □□ □□ □□

Schulman et al. (2017) □ PPO □ clip ratio ε □ □□ □□□□ □□□□ □□□□ □□□ on-policy RL □□□□□□. □ □□□ PPO □ □□□ RAG □□□ □□□ □□□ □□□□ 5,636 □□□□□ □□ Actor-Critic □□□ □□□□. □□ □□□ □□ □□□ RL □□□ Dense Passage Retrieval □ online fine-tuning □ □□ □□□ □□□ (ColBERT-PRF □), Triple-Hybrid □□□ □□□ □ □□□ □□ □□.

4 □ □□□ QA □□ □ JKSCI 2025 □□□

4.1 □□□ QA □ □□ □□

□□□ gold □□ □□ □□ □□□ □□□ ("□□□, □□□, □□□") □ □□□□, LLM □ □□□ □□□ "□□□ □□, □□□ □□, □□□ □□□ □□□□□" □□□ □□□□. □□ □□ □□ F1 □ □□ □□·□□□ □□ □□□ gold/prediction □□ □ □□□ □□□ □□□□ □□□. □ □□□ □ □□□ F1_strict (□□ □□ □□) □ F1_substring (□□ □□ □□□□ □) □ □□ □□□ □□□□□ (Ch.6 §1.3).

4.2 JKSCI 2025 □□ □□□ □□

□□ □□□ R-DWA F1 0.86, EM 0.78, Faithfulness 0.89 □ □□□□□. □ □□□ □□ □□□□□ R-DWA F1_strict □ 0.137, F1_substring □ 0.448 □ □□□□, □□ □□□ Faithfulness 0.89 □ □ □□ □□ 0.835 □ 94% □□ □□ LLM □□ □□□ □□□□ □□□□□. F1 □ □□□ □□ □□ □□ (□□□, □□□ strict/loose) □□□ □□□□ (Ch.6 §3.3 □□), □ □□□ □ □□ □□□ □□□ □□·□□ □□□□ □□□□.

▸ [□□ 2-1] R-DWA vs L-DWA — 3-□□ □□

>

▸ □□. □□ Triple-Hybrid □□□□□□ □□□ □□ □□ □□ R-DWA □ □□□ □ if-else □□ + □□ □□□ □□□□ □□□□, L-DWA □ 18-dim □□ → Actor-Critic □□□(≈6K □□□□) → Dirichlet □□□ □ □□□□ □□□ □□□ □□□□.

>

▸ □□ (□□ □□, □□ retrieval). "□□□□□ 55 □ □□ □□□?"

>

▸ | □□ | R-DWA | L-DWA |

▸ |---|---|---|

▸ | □□ □□ | (2, 1, 1) | (2, 1, 1) + 15-dim extras |

▸ | □□□ □□ | □□ □□□ + λ □□ | π_θ(s) Dirichlet mean |

▸ | □□□ (α, β, γ) | (0.15, 0.25, 0.60) (□□ □□□) | (0.12, 0.23, 0.65) (□□ □□) |

▸ | □ □□□□ F1_{strict} | 0.137 | 0.167 (+21.9%) |

>

▸ □□. □□ □□(R)□ □□□□□□ □□□, □□ □□(L)□ □□□□□□□ □□□. □ □□□ □ □ □□ □□□□ □□□□ Oracle □ 96% □□□ □□□□ □□(§3.2).

5 □ □ □□□ □□□ □□

□ □□□ □□ □ □□□ □□ □□□ □□ □□.

□	□□ □□	□ □□
□□ □□ □□	Hybrid (2-way), GraphRAG (1+□□□□)	Triple-Hybrid (Vector+Graph+Ontology)
□□□ □□	□□ / □□□ / □□	MDP + PPO □□□ □□ □□□
□□ □□□	□□ F1/EM	F1_strict + F1_substring □ □ □□

R-DWA (□□ JKSCI) □ Triple-Hybrid □ □□□□□□ □□□, □ □□ □□□ Triple-Hybrid □ □□ □□□□ □ (L-DWA) □ □□□□. □□ □□ □□□□□□□ L-DWA □ R-DWA □□ F1_strict +21.9%, Conditional F1_substring +131%□ □□□□, Oracle (per-query argmax) □□ □□□ 99% □□□ □□ □□ (Ch.6 §3).

□□ □ □□□ domain transfer □□□ □ L-DWA □□□ □□□ intent □□□ □□□□ □□ □□ □□□□ naive □□ □ Vector-only □□ □□□□ □□□ □□□□. □□ GraphRAG (English-centric) □ □ □□ (Korean-centric) □ "□□□ □□ □□□ □□□ □□" □□ □□ □□□ □□□□, □□□/□□□□ joint training □ □□□ future work □ □□□□ (Ch.7 §2).

□III□ Triple-Hybrid RAG □□□□ (v5 □□□)

□ □□ Triple-Hybrid RAG □□□□□□ □□□ □ □□ □□ — Vector, Graph, Ontology — □ □□ □ □□ □□□ □□□□. Ch.4 □ R-DWA □ Ch.5 □ L-DWA □ □□ □ □□□□ □□□ □□□□, □ □□ □□□ □□□□ □□□□□□ □□□□.

1 □ □□ □□□□□

□□□ □□□ □□ q □ □□ 7 □□□ □□ □□ a □ □□□□.

```
(1) q
(2)   → QueryAnalyzer → intent (entities, relations, constraints, density)
(3)   → DWA (R-DWA or L-DWA) → (α, β, γ) ∈ Δ³
(4)   → {VectorStore.search, GraphStore.search, OntologyStore.search} → (V, G, O)
(5)   → merge_contexts(V, G, O; α, β, γ) → context string
(6)   → LLM(prompt + context) → answer a
(7) a
```

□□ (2)~(3)□ □□□ □□ □□, □ □□□ □□ □□□□. □□ (4)~(6)□ □□□□ □□ □□□□ □□-□□-□□ □□□□.

LLM □ gpt-4o-mini (temperature=0.0, max_tokens=500) □□ □□□□.

2 □ □ □□ □□

2.1 Vector □□ (α)

□□. □□ □□□□ OpenAI text-embedding-3-small (1536-dim) □□□□ FAISS IndexFlatIP (L2 □□□ □ □□ = □□□ □□□)□ □□□□, □□ □□□□ top-k (□ □□ k=3) □□□ □□□□.

□□. src/rag/vector_store.py. Embedder □ □□ □□□ □□□□□ □□□□ □□ □□□□□□ □□□ hash embedder □ □□ (181 □ test □ 8 □□ vector store □□).

□□ □□ □□□□□□ □□. 2,542 documents (577 □□□□□ + 60 □□ □□ + 1,505 □□ □□ + 400 □□□□ □□). □□ □□□□ □□ pass □ □□ (~10 □, $0.005).

□□. □□ □□□□ □□□, □□ simple □□ □□□□ □□□ recall.

□□. □□ □□□ □□□□ □□ □□□ □□□□ □□□□ multi-hop / □□□ □□□□ □□.

2.2 Graph □□ (β)

□□. □□□ (□□, □□, □□, □□□□) □ □□ (□□, □□, □□, □□, □□) □ NetworkX MultiDiGraph □ □ □□□, □□ □□□□ □□ □□□ □□ □ BFS (max_depth=3)□ □□ □□□ □□□□. □ □□□ "A --[□□]--> B" □□□ □□□□ □□□□□.

□□. src/rag/graph_store.py. □□ □□ □□□□ □□:

- Nodes: 2,542 (60 Department + 577 Professor + 1,505 Course + 400 Project)
- Edges: ≥ 3,000 (□□ 577 + □□ ≤ 1,505 + □□ ≥ 1,700 + □□ 1,158 + □□ ≥ 800)

□□. □□□ □□□□□ □□□□ multi-hop □□ □□.

□□. □□□ □□□ □□ □□□ □□□□ □□□ □□□ □□□ □□ □□ □ □□□ □ fallback □□.

2.3 Ontology □□ (γ)

□□. OWL 2 □□□ □□ (Person → Professor → FullProfessor/AdjunctProfessor) □ □□ (hasAge, belongsTo, teaches), □□ □□ (age ≤ threshold) □ Owlready2 □ □□. HermiT □□□ □□ □ □□ □□□ □□□ □□□□, □□□ □ □□ □□ fallback□□ □□ □□ □□.

□□. src/rag/ontology_store.py. check_constraint(entity, phrase) □ "40 □ □□", "30 □ □□" □ □□□ □□ □□□ □□□ □□□□ □□ □□□□ age □□□ □□.

□□. □□□ □□ (conditional query) □ □□·□□ □□□ □□□ □□.

□□. □□□ □□ □□ □□□ □□, □□ □□□ □□ □□□ □□□□□ □□ □□ □□ □□□ □□.

3 □ QueryAnalyzer

□□. □□□ □□□□ □□(entities), □□(relations), □□(constraints) □ □□□·□□□□ □□□□, □ density □□ s_e, s_r, s_c ∈ [0, 1] □ □□ □□(simple/multi_hop/conditional) □ □□.

□□ □□.

```
@dataclass
class QueryIntent:
    query_type: Literal["simple", "multi_hop", "conditional"]
    entities: list[str]
    relations: list[str]
    constraints: list[str]
```

```
    complexity_score: float
    density: tuple[float, float, float]  # (s_e, s_r, s_c)
```

□□ □□ □□.

- constraints □□□□ □□□ → conditional
- □□□ 2 □ □□ □□ □□ 2 □ □□ → multi_hop
- □ □ → simple

▸ [□□ 3-1] □□ □□ (Density Signals) — □□ · □□ · □□

>

▸ □□. □□ □□ □□ □□□□ □ □□□□ □□□□ □ □□ □□ (c_e, c_r, c_c). □□ □□ □ □□(entity), □□(relation), □□(constraint) □□□ □□.

>

▸ □□ □□.

▸ c_e = |{e ∈ q : e ∈ $\mathcal{E}$}|, c_r = |{r ∈ q : r ∈ $\mathcal{R}$}|, c_c = |{t ∈ q : t ∈ $\mathcal{C}$}|

>

▸ □□ □ □□ ·□□·□□ □□□ □□, $\mathcal{R}$ □ □□·□□ □□ □□ □□, $\mathcal{C}$ □ "N □ □□" □ □□ □□ □□□ □□.

>

▸ □□. □□ "□□□□□ □□ 55 □ □□ □□□?":

>

▸ | □□ | □□ □□ | □ |

▸ |---|---|---|

▸ | c_e | "□□□□□", "□□" | 2 |

▸ | c_r | "□□" | 1 |

▸ | c_c | "55 □ □□" | 1 |

>

▸ □□. (c_e, c_r, c_c) = (2, 1, 1). R-DWA □ □ □□□ □□ "□□□ □□□□ □□□ Ontology □□□(γ)□ □□□"□ □□ (Ch.IV). L-DWA □ □□ □□□ 18-dim state □ □□□□ □□□□□ □□□□ □□ (Ch.V).

□□ □□. §V.2 □ BERT multi-label intent classifier (src/intent/bert_classifier.py) □ multi-label □□ □□□ □□□□ L-DWA □ state (intent_logits 3-dim) □ □□□□ □ □□. □ □□□ □ □□□□ □□□ □□□ intent_logits=(0,0,0) □□ □□ (Ch.6 §5.2 □□).

4 □ Score Fusion: merge_contexts

□ □□□ □□□ □□□ □□□ □□ □□□□ □□□ context □□□□ □□□□. top_k=3, budget = top_k × 3 = 9 □□ □□:

```
n_v = max(1, round(9 · α))
n_g = max(1, round(9 · β))
n_o = max(1, round(9 · γ))

context = [
    f"[Vector(α={α:.2f})]\n" + "\n".join(V[:n_v]),
    f"[Graph(β={β:.2f})]\n"  + "\n".join(G[:n_g]),
    f"[Ontology(γ={γ:.2f})]\n" + "\n".join(O[:n_o]),
]
```

□□ □□. □□□□ □□ LLM □□□ □□ □□ □ □□□□□ □□ □□□ □□□□. □ □□□ □ □□□ □□ 1 □□□ □□ (max(1, ·)) □□, □ □□□□ □□□ □ □□ □□□ □□□□□ □□ □□□□.

□□. src/rag/triple_hybrid_rag.py::merge_contexts (□□ □□ □□, Ch.6 □ offline cache builder □ evaluate_rerun □□ □□ □□).

- [□□ 3-2] □□□ □□□ Δ^3 — □□ · □□ · □□

>

- □□. 3 □ □□ □□□ □□ (α, β, γ) □ □□□□ □□ □□ □□. "□□ □□□ 0 □□, □□ □□□ 1".

>

- □□ □□.
- $\Delta^3 = \{(\alpha, \beta, \gamma) \in \mathbb{R}^3 :\ \alpha + \beta + \gamma = 1,\ \alpha, \beta, \gamma \geq 0\}$

>

- □□. Δ^3 □ □□ □ □:

>

- | □□ | (α, β, γ) | □□ |
- |---|---|---|

▸ | Vector-only | (1.0, 0.0, 0.0) | □□□□ □□, Graph/Ontology □□ |

▸ | Uniform | (0.33, 0.33, 0.33) | □ □□ □□ □□□ |

▸ | Ontology-heavy | (0.1, 0.2, 0.7) | □□□ □□□ □□□□ □□ |

>

▸ □□. Δ^3 □□ □ □□ "□□ □□□ □□□ □□□" □ □□ □□□ □□. R-DWA □ □□ □□, L-DWA □ □□ □□ □ □□ □□□□ □□□□. Offline cache □ Δ^3 □□ 66 □ □□ □□□ (10-grid □□□ □□□) □□ □□□□ PPO □□□ □□ □□□ 85% □□□□ (§VI.10).

5 □ Prompt Template

□□ □□□ □□ □□□ □□ □□□□□ □□□□, □□□□ □□□ □□ □ □□ □□□ □□□□ □□□ □□□□:

```
□□ □□□□□ □□□□ □□□ □□□□ □□□□□.
□□□□□□ □□ □□ □ □□ □□□□ '□□□ □□ □ □□□□'□□ □□□□.
Graph □□ □□(A --[□□]--> B)□ □□□□ □□□ □□□□□.
□□□ □ □□□□□ □□□□ □□□□□.

Context:
{context}

Question: {query}

Answer:
```

□□ □□.

5. "□□□ □ □□□□□ □□□□" — □□ □□ □□, F1 □□ □□
6. "Graph □□ □□ □□" — LLM □ □□ □□□ □□□□ □□□□□□ □□□
7. □□ □ "□□□ □□ □ □□□□" — □□ □□

6 □ Pluggable DWA Interface

R-DWA (Ch.4) □ L-DWA (Ch.5) □ □□ □□ □□ □□□□□□ □□□□:

```
class BaseDWA(ABC):
    @abstractmethod
    def compute(self, query: str, intent: QueryIntent) -> DWAWeights:
        """(α, β, γ) on the 3-simplex."""
```

DWAWeights □ simplex □□ (α + β + γ = 1, □ ∈ [0, 1]) □ __post_init__ □□ □□□□

frozen dataclass. □ □□□ □□□ TripleHybridRAG.query() □□□

R-DWA/L-DWA/Uniform/Vector-only □ □□ □□□ □□ □□ □□.

□ □□□□□□ □□ □□ □□ "Quadruple-Hybrid" (Audio □□ □□) □ □□ □ BaseDWA □ □□ □□□ merge_contexts □ □□ □□□ □□□□□ □□□ □□□□□. □ □□□ 3-simplex □ □□□□.

□ □ □□ □□□ □□

Ch.3 □ □ □□□ □□□ □□. R-DWA (Ch.4) □ L-DWA (Ch.5) □ □ □□□ □□ □□ □□□ □□□□, Ch.6 □□ □ □□□ □□ □□□□□ □□ □□□□. □□□ □□ (□ □□□ □□, merge_contexts, Prompt) □ □ □ □□□ □□□□ □□□□ □□□□.

□IV□ □□ □□ □□ □□□ □□□□ (R-DWA)

□ □□ □□ □□ (Shin & Moon, 2025, JKSCI)□ Rule-based Dynamic Weighting Algorithm □ □ □□□ □□□□□□□□ □□□□. Ch.5 □ L-DWA □ □□□□ □□□□, Ch.6 □ □□□ □□ □□□ ·□□ □□ □□□□ R-DWA □ □□□□ □□ □□ □□□□□□□ □□□.

1 □ R-DWA □□ □□

R-DWA □ □ □□□ □□□ (α, β, γ) □ □□□□.

Stage 1 — □□ □□ → □□ □□□.

QueryAnalyzer(Ch.3 §3)□ □□□ □□□ □□ □□ □□□ □□ □□□□ □□□□ (Table 4-1):

□□	**α (Vector)**	**β (Graph)**	**γ (Ontology)**
simple	0.6	0.2	0.2
multi_hop	0.2	0.6	0.2
conditional	0.2	0.2	0.6

Stage 2 — □□ □□ □□ □□ (□□ 4-2 ~ 4-4).

□□ □□ s_r (□□ □□), s_c (□□ □□) □ □□ □□□□ □□ □□ (λ=0.3, grid search □ □□):

```
α' = α_base × (1 − λ · (s_r + s_c) / 2)
β' = β_base + λ · s_r · (1 − β_base)
γ' = γ_base + λ · s_c · (1 − γ_base)
```

Stage 3 — □□□ (□□ 4-5).

```
(α, β, γ) = (α', β', γ') / (α' + β' + γ')
```

▸ [□□ 4-1] R-DWA □□ □ □□ □□□□ — □□ · □□ · □□

>

▸ □□. □□ □□ □□ → □□ □□□ □□□ → □□ □□□ □□ □□ → □□□, □ 3□□□ (α, β, γ) □ □□□□ □□ □□ □□□□. □□□ □□ □□□ · API □□ □□ □□ □□ □□.

>

▸ □□ — "□□□□□ □□ 55 □ □□ □□□?".

>

▸ | □□ | □□ | □ |

▸ |---|---|---|

▸ | (1) QueryAnalyzer | type="conditional", (s_e, s_r, s_c) = (0.40, 0.20, 0.20) | |

▸ | (2) □□ □□□ (Table 4-1) | conditional → | α_{base}=0.20, β_{base}=0.20, γ_{base}=0.60 |

▸ | (3a) α' | 0.20 × (1 − 0.3 × (0.20 + 0.20) / 2) | 0.188 |

▸ | (3b) β' | 0.20 + 0.3 × 0.20 × 0.80 | 0.248 |

▸ | (3c) γ' | 0.60 + 0.3 × 0.20 × 0.40 | 0.624 |

▸ | (4) □□□ | □□ 1.060 □□ □□ | (0.178, 0.234, 0.588) |

>

▸ □□. Conditional □□□ □□ □□□ (0.2, 0.2, 0.6) □□ □□ □□ s_c=0.20 □ γ □ 0.6 → 0.588 □ □□□□ □□(□□□ □□) □□□□ □□□ Ontology □□□ □□□. R-DWA □ □□□ □ "□□□ + □□ □□" □□□ □□ — Ch.5 L-DWA □ □ 3-□□ □□□ □□□ Dirichlet □□□ □□□□.

2 □ R-DWA □ □□

8. □□□·□□□. □□□ □□ □□□ □□ API □□ □□ □□ □□ □□.
9. □□ □□□. □□ □□□ □□□ □□□ □□□□ □□ □□ □□□□ □□ □□.
10. □□□. □□□ □□□□□ □□ □□□ □□ □□□ □□ □□.

3 □ R-DWA □ □□□ □□

3.1 □□ □□□ □□□□ □□□ □□

Table 4-1 □ □□□ □□ (simple→α, multi_hop→β, conditional→γ) □ □□□ □□□ □□□ □□□ □□ □□□□. □ □□ Ch.6 §3.3 □ □□ □□□ □□□, □□ □□ □□□□□ per-query argmax Oracle □□ □□□□ (α=0.11, β=0.16, γ=0.73) □□ □□□□ □□□□ Ontology □ □□□□□. □ R-DWA □ simple→α □□□□□ □□ □□ □□□ □□□□ □□□.

3.2 □□ □□ λ □ □□ □□□□□□□□

Stage 2 □ λ=0.3 □ □□ □□ □□□ □□ □□□ □□□□. □□□ □□ (□□, □□□ □, □□ □□ □□ □) □ □□□□ □□□.

3.3 □□□ □□□ □□ □□ □□

Conditional □□ □□□□□ (□□ □□ / □□□ □□ / □□ □□) □ □□ □□□ □□□□, R-DWA □ □□ □□□ (0.2, 0.2, 0.6) □□ □□□□. Ch.6 §4 □ per-type □□□ conditional □□ L-DWA □ F1_substring +131%□ □□□□ □□, □ □□□ □□□□□ □□□□ □□□□.

3.4 □□□ □ □□ □□

R-DWA □ Table 4-1 □ λ □ □□ □□ □□□□□ □□□□□. HotpotQA/MuSiQue □ □ □□□□□ □□□ □□ □□ □ □□□□□ Vector-only □□ □□□ □□□ □□□ Ch.6 §7 □□ □□□□□ (HotpotQA F1_strict R-DWA 0.097 vs Vector-only 0.102).

4 □ R-DWA □□ □ □□

□□. src/dwa/rdwa.py::RuleBasedDWA(lambda_=0.3).

□□. src/intent/rule_based.py::RuleBasedIntent (□□□ □□□ □□ QueryAnalyzer).

□□□. tests/test_dwa_rdwa.py 9 □ □□ □□□ (□ □□ □□ □□ □□).

5 □ R-DWA □ □ □□ □□□ □□

5.1 □□ (5,000 QA, □□ □□ □□□)

□□	**R-DWA** □□□
F1_strict	0.137 ± 0.185
F1_substring	0.450 ± 0.445
EM	0.000
Faithfulness	0.835 ± 0.354
Total R	0.199 ± 0.120
□□ latency	1.19s

□□ □□ □□□: (α=0.25, β=0.45, γ=0.30). □ R-DWA □ Graph □□□ □□ (multi-hop □□□□ □□). □□ Oracle □ Ontology □□□□ □□ □□□□.

5.2 □□ □□□

Type	n	F1_strict	F1_substring	Faith
simple	2,000	0.153	0.837	0.928
multi_hop	1,750	0.017	0.261	0.728
conditional	1,250	0.019	0.090	0.806

Conditional □ multi_hop □□ F1 □ □□□ □□. □□ (i) □ □□□ gold □□ □□□□□ □□ □□ □□□□□ LLM □□□ □□□□ □□ □□□ □□□, (ii) R-DWA □ □□ □□□□ □□ □□□ □□□□ □□□□. Ch.5 L-DWA □ □ □□□ □□□ □□□□□□ □ □□□ □□ □□□□.

5.3 □□ JKSCI □□ □□

□□	JKSCI □□	□ □□ □□	□□□
F1	0.86	0.137 (strict) / 0.450 (substring)	16% (strict) / 52% (substring)
EM	0.78	0.000	0% (□□□)
Faith	0.89	0.835	94%

Faithfulness □ 94% □□□ LLM □□ □□□ □□□□ □□□□. F1 □ □□□ □□ □□ □□□□ □□□ □□ □□□□ □□□□□ □□□□ (Ch.6 §3.3). F1_substring 0.450 □ □□ F1 0.86 □ 52% □□□□, □□ □□□ substring □□□ □□□ □□ □□ □□ □ □□□□ □□□□ (□: □□□ □□ □□ □□ □ □ □□□ □□□ □□□). □ □□ □ □ □□□ □□□□ □□ □□□□ □□□□ □□ □□□□ □□□ □□□□.

6 □ □□

R-DWA □ □□□ baseline □□□ □□□ □□□, □ □□□ Ch.5 L-DWA □ □□ □□□□□ □□□ □□□ □□□□ □ □□□ □ □□ □□ □□□□□. R-DWA □ □□□ — □□□ □□, □□ λ, □□ □ □□ □□ □□, □□□ □□ □□ — □ Ch.5 □ □□□ □□□ □□□□□□.

□V□ □□□ □□ □□□ □□□□ (L-DWA)

□ □□ □ □□□ □□ □□ □ Learned Dynamic Weighting Algorithm □ □□□□. §1 □□ MDP □□ □, §2 □□ Actor-Critic □□ □□□□, §3 □□ PPO □□ □□□□, §4 □□ □□□□ □□ □□ □□□□ □□□.

1 □ MDP □□□

Triple-Hybrid RAG □□□ □□□ □□ 1-step contextual bandit MDP □ □□□□□.

1.1 □□ (State, 18-dim)

□ □□□ □□ 18□□ □□ □□□ □□□□ (□ 5-1):

s = [density (3) | intent_logits (3) | source_stats (9) | query_meta (3)]

□□	□□	□□	□□
density	3	RuleBasedIntent	(s_e, s_r, s_c) □□□ □ □ □□
intent_logits	3	BERT classifier (□□ 0)	(logit_simple, logit_multi, logit_cond)
source_stats	9	Retrieval □□	3 □□ × (top, mean, count)
query_meta	3	Query string	(log_length_norm, n_entities_norm, has_negation_flag)

intent_logits. □ □□ □□□□□ BERT multi-label classifier □ □□□ □□□ □□

intent_logits=(0, 0, 0) □□ □□. □ □□□ □□□ Ch.6 §5.2 □ □□ □□□□ □□□. □□□ density □□

□□ L-DWA □ Oracle □ 99%□ □□□□□ BERT □ □□ □□ □□ □□ □ □□□.

source_stats. □ □□□ retrieval □□□□:

- top = □□ 1□□ □□□ □□ (Vector □ □□ 0~1, Graph/Ontology □ □□ □□ □□ 0/1 □□□)
- mean = □□ top_k □□□ □□
- count_norm = min(len(results) / top_k, 1.0)

1.2 □□ (Action)

3-simplex □□ □□ □□:

$a = (\alpha, \beta, \gamma) \in \Delta^3, \ \alpha + \beta + \gamma = 1, \alpha,\beta,\gamma \geq 0$

Actor □□□□□ Dirichlet □□□ □□ □□□□ $c = (c_\alpha, c_\beta, c_\gamma)$ □ □□□□, □□ □□□ Dirichlet(c)

□□, □□ □□□ Dirichlet mean (= $c/||c||_1$) □ □□□□.

1.3 □□ (Reward, Eq. 5-7)

$R(s, a) = 0.5 \cdot F1 + 0.3 \cdot EM + 0.2 \cdot Faithfulness - 0.1 \cdot \max(0, latency - 5.0)$

□□□ □□ □□. □ □□□□□ gold □□ □□ □□ □□□ □□□□ EM = 0□ □□□□□ □□□□ (Ch.6 §1.3). □□

□ □□ □□□

$R \approx 0.5 \cdot F1 + 0.2 \cdot Faithfulness - penalty$

□□, F1□ Faithfulness□ □□ □□□ □ □□□□. □ □□□ □ □□□ Ch.6 §2.3 □□ □□□□ □□□□.

1.4 □□-□□ □□

□□□□□□ □□ (state, action, reward) □□□ □□□□□□ GAE □ □□ □□□ □□□□□□ 1-step Bellman: A = R − V, returns = R. □ □□□ □□□ GAE machinery □ □□□□ □□ multi-turn dialog □ □□ □□□□.

▸ [□□ 5-1] MDP □□ (S, A, P, R, γ_d) — □□ · □□ · □□

>

▸ □□. PPO □□ □□□ □□□□ 5 □ □□. □ □□□ □□□:

>

▸ | □□ | □□ | □ □□ □□□ |
▸ |---|---|---|
▸ | S | □□ □□ | 18 □□ □□ (density·intent·source_stats·query_meta) |
▸ | A | □□ □□ | Δ^3 □□ □□□ (α, β, γ) |
▸ | P | □□ □□ | 1-step MDP (□□ □□ □□) ≈ □□□□ □□ |
▸ | R | □□ □□ | 0.5·F1 + 0.3·EM + 0.2·Faith − 0.1·max(0, lat−5) |
▸ | γ_d | □□ □□ | 0.99 (1-step □□□ □□ □□□) |

>

▸ □□. □ □□ □□□:
▸ 1. □□ "□□□□□ 55 □ □□ □□□?" □□ → s (18-dim) □□
▸ 2. □□ π(·|s) □ a = (0.1, 0.2, 0.7) □□□
▸ 3. □□□□□□ a □ □□□□ □□ → LLM □□ → F1=0.8, Faith=0.9 □□
▸ 4. R = 0.5·0.8 + 0.2·0.9 = 0.58 (EM=0, latency<5)
▸ 5. (s, a, R) □□□ PPO □□ □□□□

>

▸ □□. □ □□□ MDP □ □□ step □□□ value function □ □□□ □□□□, PPO □ clipping □ □□□ □□□ □□□□ (□□ 5-2 □□). □□ □□□ □□□□ 1-step bandit □□□ □□□ □□□□, PPO □□□ □□□□ □□ □□□·□□□ □□□ □□.

2 □ Actor-Critic □□□□

2.1 □□ (Figure 5-3)

```
s (18) → Linear(18→64) → Tanh → Linear(64→64) → Tanh → features (64)
                                          ↓
                    Actor head:  Linear(64→3) → Softplus + ε → Dirichlet concentrations c
                    Critic head: Linear(64→1)                → □□ □□ V(s)
```

2.2 □□□□ □

□ □□□ □□□□ □□□ □□ □□ backbone □ □□□□. □ 5,636 □□□□:

□	**in × out + bias**	□□□□
Linear (18→64)	18·64 + 64	1,216
Linear (64→64)	64·64 + 64	4,160
Actor head (64→3)	64·3 + 3	195
Critic head (64→1)	64·1 + 1	65
□□		5,636

□□ □□ □ □□ RL □□□ □□ □□□□□ □□□ ~ □□□ □□□□ □□□□, □ □□□ □□ □□ (18-dim) □ □□ □□ (3-simplex) □ □□□□ □□□ □□ □□□ □□□ □□.

2.3 □□□ □□

Orthogonal init (gain=1.0). □□ Linear □□ weight □ □□ □□□□□, bias □ 0 □□ □□. □ □□ □ □□: □□ □□□ Dirichlet □□□ ≈ (1/3, 1/3, 1/3) □ □□, R-DWA □ "□□□ prior" □ □□ □□□□ □□□ □□□□ □□. □□ (i) □□ warm-up □□ □□□ □□ □□□ □□□ □□, (ii) □□ □□□ □□□ □□□ (R-DWA baseline □□ □□□ movement □□□).

3 □ PPO □□ □□□□

3.1 □□ PPO with Clip

□□ □□:

```
L^π(θ) = −E[ min(ratio · A, clip(ratio, 1−ε, 1+ε) · A) ]
ratio = π_θ(a|s) / π_θ_old(a|s)
```

□□ □□:

```
L^V(φ) = 0.5 · E[(R − V_φ(s))²]
```

□□□□ □□□:

```
L^H(θ) = −H(π_θ)
```

□ □□ (Actor-Critic □□ backbone):

```
L(θ, φ) = L^π + c_V · L^V − c_H · H
```

▸ [□□ 5-2] PPO Clip □□ □□ — □□ · □□ · □□

>

▸ □□. □□ □□□□ □ □ □□□ □□ □□□ □□□□ □□□□ □□□ □□□ r_t(θ) □ □□(clip) □□□ □□□.

>

▸ □□ □□.
▸ L^{CLIP}(θ) = 𝔼_t[min(r_t(θ)·Â_t, clip(r_t(θ), 1−ε, 1+ε)·Â_t)]

>

▸ □ □□ ε = 0.2 (Table 5-4).

>

▸ □□. □ (s, a) □□□□ clipping □□:

>

▸ | □□ | r_t | Â_t | Clipped r_t | □□ term | □□ |
▸ |---|---|---|---|---|---|
▸ | □□□ □□ □□ □□ | 1.1 | +0.5 | 1.1 | +0.55 | □□ □□□□ |
▸ | □□□ □□ □□ □□ | 1.5 | +0.5 | 1.2 | +0.60 | clip □□ |
▸ | □□□ □□ □□ □□ | 0.9 | −0.3 | 0.9 | −0.27 | □□ □□□□ |
▸ | □□□ □□ □□ □□ | 0.5 | −0.3 | 0.8 | −0.24 | clip □□ |

>

▸ □□. clip □□ vanilla PG □ □ step □ □ □□□□□ □□ □□ □□ □□ □ □ □□. PPO clip □ "□ □□ ±20% □□ □□□ □ □" □ □□□□ □□ □□□ □□. □ □□ 3-seed □□□□ R_{late} = 0.215 ± 0.002 □ □□□ □□□□ clip □□□□ (§3.5).

3.2 □□□□□□□ (Table 5-4)

□□□□	□	□□
learning rate	3e-4	Adam □□, RL □□ □□
gamma (□□)	0.99	1-step □□ □□ □□
GAE lambda	0.95	□□
clip ratio ε	0.2	Schulman et al. (2017)
value coef c_V	0.5	□□
entropy coef c_H	0.01	□□ □□

max grad norm	0.5	□□□
rollout □□	32	□ episode □
minibatch □□	8	rollout □
update epochs	4	rollout □
□ episodes	10,000	seed □ (Ch.6 §5)

3.3 Advantage □□□

□ rollout □□□ advantage □ □□ 0, std 1 □ □□□□□ □□ □□ reward scale □□ □□ □□□□ □□ □□.

3.4 □□□

PPO □□□ 3 seed × 10,000 episode:

- seed 42: Total R 0.1988 → 0.2145 (Δ +0.016)
- seed 123: Total R 0.1970 → 0.2153 (Δ +0.018)
- seed 999: Total R 0.1964 → 0.2146 (Δ +0.018)

3 seeds □ □□ R □□□□ 0.0016 (0.7%) — □□ □□□□□ □□ □□□□ □□□□.

3.5 □□ □□ □□

□□ seed □□ episode ~2,000 □□ □□□ R □□ □ R ≈ 0.21 □□ □□ □□□□ plateau □ □□□□.

Policy entropy H □ ep 1 □ −0.8 □□ ep 10,000 □ −2.7 □□ □□ □□□□ □□ □□ → □□□ □□□ □□ □□□ □□□.

4 □ □□□□ □□ □□

4.1 □□ □□

3 seed × 10,000 episode × 32 rollout = 960,000 □□ □□ □ on-policy □□□ □□□□.

gpt-4o-mini 1 □□ □□ $0.0003 □□ $288, □ □□ □ 50 □□ □□ (□□ □□ □□) □ □□□□.

4.2 □□ □□

□ □□□ 5,000 □□ × 66 □□ □□□ = 330,000 □□□ □ □□□ SQLite □ □□ □□□□.

□□□. □□□ □□ Δ=0.1 □□□□ α+β+γ=1 □ □□□□ □□ □□□ □:

```
(a_int, b_int, g_int) ∈ {0,1,...,10}³, a_int + b_int + g_int = 10
□ |S_Δ| = C(12, 2) = 66
```

Schema.

```
CREATE TABLE rewards (
    query_id TEXT, alpha_int INT, beta_int INT, gamma_int INT,
    f1 REAL, em REAL, faithfulness REAL, latency REAL,
    PRIMARY KEY (query_id, alpha_int, beta_int, gamma_int)
);
```

□□. □□ □ Dirichlet □□□ □□ □□□□ □□ □□□ □□□ □ O(1) □□ □□. □□ □□□ Δ/2 = 0.05 □□.

4.3 □□ □□/□□

□□ □□ □□ (330K □□□, 10 □□ □□):

- □□ wall-clock: 14 □□ (□□ □□ □□ □□)
- □□ □□: $33 (OpenAI invoice □□)
- □□ □□: 16.8 MB

4.4 □□ □□ □□ (□□ □□)

□□	□□	□□ **(3 seeds)**
On-policy (□□ □□)	$288	≥ 50 □□
□ □□ offline cache	$33 (□□) + $0 (□□)	14h (□□) + 1h (□□) = 15h
□□□	−85%	−70% (□□ seed □□ □□□□□ □ □ □□)

Amortization. □□□ □□ □□ □ □□ seed, □□ ablation, □□ policy □□□ □□□□□. N seed □□ □ □□□ □□□ $33 (□□□ $0), on-policy □ $96·N.

4.5 □□

src/utils/offline_cache.py::OfflineCache. ThreadPoolExecutor n_workers=10 □□ □□ □□, SQLite WAL □□, put_many □□ □□□□ □□□ □□. scripts/build_cache.py □ □□ □□ □□□□□, scripts/cross_build_cache.py □ □□ □□□ (HotpotQA □) □.

5 □ □□ □□□ □□

5.1 StateProvider □ RewardFn □□

PPOTrainer □ □ callable □ □□□□:

- state_provider(query_index: int) → State
- reward_fn(query_index: int, weights: DWAWeights) → float

□□□ □□ □□□□, □□ □ pre-compute state pickle □ □□□□□ cache □□□ □□ □ □□□ □□□ □□ □□.

5.2 BERT intent classifier □ □ □□

src/intent/bert_classifier.py::BertIntentClassifier □ klue/bert-base □□ multi-label □□□□ □□□□ □□ (3 labels = simple/multi_hop/conditional, BCE loss, tokenizer + model DI). □ □□ □□□□□ □□ □□ □□□□ □□□ □□ □ □□ intent_logits=(0, 0, 0) □ □□□□□.

□ □□□□ L-DWA □ Oracle □ 99.4%□ □□□ (Ch.6 §3) □ density + source_stats + query_meta □ 15 □□□□□□ □□ □□□ □□□□ □□□□. BERT □□ □ □□ □□ □□ □ Ch.7 §2 □ □□ □ □ □□□□.

5.3 □□ □□ □□□□□

```
1. □□□ □□ (dataset_generator.py seed=42)
2. VectorStore □□ (OpenAI □□□ × 2542 □□, ~10 □, $0.005)
3. 5000 □□ × 66 □□□ □□ □□ □□ (14 □□, $33)
4. For each seed ∈ {42, 123, 999}:
   4a. 5000 □□ state pre-compute (~20 □, □ seed □)
   4b. PPO □□ 10,000 episodes (GPU 16 □, $0)
   4c. final.pt □□□□□ □□
5. L-DWA □□ (fresh LLM, ~10 □/□□, ~$3 per □□)
```

□ □□: □ 15 □□, $38 (3-seed □□ ablation □□).

6 □ □□

□ □□ MDP □□□, 5,636 □□□□□ □□ Actor-Critic, PPO □□ □□□, □□□ □□ □□ 85% □□ □□□□ □□□ □□□□ L-DWA □□□□□ □ □□□□□. □ □□□□□ □□□ Ch.6 □ □□□ R-DWA □□ F1_strict +21.9% □ Oracle 99.4% □□□ □□□□.

□VI□ □□ □ □□ (v5)

□ □□ □ □□□ □□ □□□ □□□□. §1 □□ □□, §2 □□□□ □□ □□ □□ □□, §3 □□ □□ □□ (3-seed □□), §4 □□ □□□ □□, §5 PPO □□ □□□, §6 □□ □□, §7 □□ □□□ □□, §8 □□ □ □□, §9 □□.

1 □ □□ □□ □ □□ □□

1.1 □□□□·□□□□□

- GPU: NVIDIA GeForce RTX 4090 (24 GB VRAM)
- Host: Windows 11 Pro + Docker Desktop (WSL2)
- Base image: nvidia/cuda:12.1.0-cudnn8-devel-ubuntu22.04
- Python 3.10.12 · PyTorch 2.1.2+cu121 · FAISS 1.7.4 · Owlready2 0.46 · Transformers 4.41.2 · numpy 1.26.4 (PyTorch 2.1 □□ □)

1.2 LLM, □□□, Retrieval

- LLM: gpt-4o-mini (2024-07-18 □□□), temperature=0.0, max_tokens=500
- Embedding: text-embedding-3-small (1536-dim)
- Retrieval top-k = 3, Graph BFS depth = 3

1.3 □□ □□ □□ (□ □□□ □□ □□)

□ □□ F1 □ □□ □□:

□□	□□	□□
F1_strict	□□□ □□ □□ + □□□ □□ + NFC □□□ □ □□ □□ F1	□□□ □ □□
F1_substring	Gold □ □□□ □□□ □□□□ prediction □ □□□□□□ □□□□ □ □ □□ F1	□□ □□ loose □□
F1_char	□□□ □ □□ 3-gram □□□ F1	□□·□□·□□□ □□, □ 3 □ □□

EM = 0 □□ (sentence □□□□ □□). Gold □□ "□□□, □□□, □□□" □ □□ □□ □□ □□ □□ □□ sentence-style □□□□□ LLM □ "□□□ □□, □□□ □□, □□□ □□□ □□□□□" □□□ □□□ □□□□ □□□ □. □□ □□ □□ □□□ □□□□□ □□□□□□. □ □□ §3.4 □□ □□□ list-style □□□□ (PROMPT_TEMPLATE_LIST) □ □ □□□ □□□□ EM ≈ 0.39 □□ □□□□.

Faithfulness (2-branch): RAGAS □□ □□. □□ □□□ □□ □ □□□ □□□□. (i) □□□ (□□□ □□): □ □ □□ any-token □□ (□ □□). (ii) □□□□ (□□ + □□□ □□): □ □□ □□□ □□ claim □□ □□. □□□ □ □□ □□□ □□□ □ □□□ □□□□□ □□ □□ □□□□ (§3.5 □□).

▸ [□□ 6-1] F1_{strict} — □□ □□□□ F1
>

▸ □□. □□□ □□□ □□□ □□□ (NFC → □□□ → □□ □□ → □□□ □□ → □□ □□ → □□□ □□) □ □□ □□ □□ □□□□ F1 □□.

>

▸ □□.

▸ P = |T_p ∩ T_g| / |T_p|, R = |T_p ∩ T_g| / |T_g|, F1 = 2PR / (P + R)

>

▸ □□. Gold: "□□□, □□□, □□□" (T_g = {□□□, □□□, □□□})

>

▸ | Pred | T_p | F1_{strict} |

▸ |---|---|---|

▸ | □□□: "□□□ □□□ □□□ □□, □□□ □□□ □□□□□" | {□□□, □□, □□□, □□□, □□} | 0.75 |

▸ | □□□□: "□□□, □□□, □□□" | {□□□, □□□, □□□} | 1.00 |

>

▸ □□. □□ □□□□□ □□□ □□ F1 □ □□□. □ □□□ □ □□□□ □□□□ F1_{substring}, F1_{char} □ □□ □□ □ □□.

▸ [□□ 6-2] F1_{substring} — □□□ □□ □□ F1

>

▸ □□. Gold □□ □□□ □□□□ □□ □□(item) □ prediction □□□□ □□□□□□ □□□□ □□.

>

▸ □□. G = {g_i} □ gold □□ □□ □□, h = |{g_i: g_i ⊆ p}|, Recall_{sub} = h/|G|, Precision_{sub} = h / max(n_{chunk}, |G|).

>

▸ □□. Gold: "□□□, □□□, □□□" (|G|=3)

>

▸ | Pred | h | F1_{substring} |

▸ |---|---|---|

▸ | "□□□ □□□ □□□ □□, □□□ □□□ □□□□□" | 3 | 1.00 |

▸ | "□□□□ □□□□□" | 1 | 0.50 |

▸ | "□□□□ □□□□□" | 0 | 0.00 |

>

▸ □□. □□□ □□□□ JKSCI 2025 □ "loose" F1 (0.86) □ □□ □□□ □□ □□. □ □□ 5,000 QA □ □□□ 0.450 (R-DWA) / 0.488 (L-DWA) □ journal □□□ 52% / 57% □□ (§3.3).

▸ [□□ 6-3] F1_{char} — □□ 3-gram F1

>

▸ □□. Pred·Gold □□□□ □□ 3-gram □□ □□ □□ □ F1 □□. □□·□□·□□□□ □□□ □□ □□ □□ □□.

>

▸ □□. C(x) = {x[i:i+3] : 0 ≤ i ≤ |x|−3}. T_p = C(p̂), T_g = C(ĝ) (p̂, ĝ □ □□□·□□ □□ □□□). F1 = 2·|T_p ∩ T_g| / (|T_p| + |T_g|) .

>

▸ □□. Gold "□□□": C = {□□□} (1 □)

>

▸ | Pred | C(Pred) | □□ | F1_{char} |

▸ |---|---|---|---|

▸ | "□□□" | {□□□} | 1 | 1.00 |

▸ | "□□□□□" | {□□□, □□□, □□□} | 1 | 0.50 |

▸ | "□□□" | {□□□} | 0 | 0.00 |

>

▸ □□. □□ □□ □□ → □□ □□ (□: "□□□□" vs "□□□") □□ □□. □ □□□ F1_{strict} □ F1_{substring} □□□ □ 3 □□□□ □□.

▸ [□□ 6-4] Faithfulness (2-branch) — □□ · □□ · □□

>

▸ □□. RAGAS □□ □□ □□□. □□ □□□ □□ □ □□:

▸ - □□□ (□□□ □□): claims = sentence_split(answer), □ □□□ □□ ≥ 2 □□ □□□□ □□□□ □ □□□ □□.

▸ - □□□□ (□□ + □□□ □□): claims = comma_split(answer), □ □□□ □□ □□□□□ □□ □□ □□ □□ □ □□□□□ □□□ □□.

>

▸ □□. Context = "□□□ □□□ □□□ □□□ □□□ □□□□" □ □□:

>

▸ | Answer | □□ | □□/□ | Faith |

▸ |---|---|---|---|

▸ | "□□□ □□□ □□□ □□□ □□□□□." | □□□ | 1/1 | 1.00 |

▸ | "□□□, □□□, □□□" | □□□□ | 2/3 | 0.67 |

▸ | "□□□, □□□" | □□□□ | 2/2 | 1.00 |

>

▸ □□. □□□ □□□ □□□ □□ □□ □□(hallucination) □□□ □ □□. list-prompt □□□ faithfulness □ sentence-prompt □□ □□ □□□□, □□ □□□□ □□ □□□ □ □□□ □□□□ □□□□ □ □ □□□ □□ (§3.4 □□).

Total Reward R:

```
R = 0.5 · F1_strict + 0.3 · EM + 0.2 · Faithfulness − 0.1 · max(0, latency − 5.0)
```

PPO □□□ □□ □□. Ch.5 □ 5-7 □ □□.

1.4 □□□ □□ □□ □□

□ □□ □□ □□, □□ f1_score □ □□·□□□□ □□□□ □□□□ □□ gold "□□□," □ prediction "□□□" □ □□□□ □□ □□□ □□□□□. □□ □□ (src/eval/metrics.py::normalize_korean □ _PUNCT_RE □□) □ □□ F1_strict □□ □ 90% □□□□□. □ □□□ §3 □□ □□ □□□ □□□ □□□ □ □□□□. □□□ 330K □□ □□□ □□ □□ □□□□ □□□□□□□□ PPO □□□ □□□ F1 □□□ □□□□□□ □□ □□□ □□□□ □□ □ □□ □□ □□ □ Ch.6 §5.2 □□ □□□□.

1.5 □□□□

- □□ □□ □□□ (□ □ §2~§6): dataset_generator.py seed=42. 60 □□ / 577 □□ / 1,505 □□ / 400 □□□□ / 1,158 □□ / 2,542 □□.
- Gold QA: 5,000 □ (simple 2,000 / multi_hop 1,750 / conditional 1,250), gold_qa_5000.json.
- □□ □□□ (§7): HotpotQA Hard 300, MuSiQue Dev 300, PubMedQA Pharma 300.

1.6 □□□ □□

□□ PPO □□ □ □□□ □□□ {42, 123, 999} 3 □□. □□□ □□ ± □□□□□ □□.

2 □ □□□□ □□ □□ (M5 □□)

2.1 □□ □□

□□	□
□□ □□□	330,000
□□ □	5,000
□□ □□ □□	16.8 MB
□ wall-clock	14 □□ (10 □□ □□)
□ □□	$33 (OpenAI invoice)

2.2 □□ □□ □□

PPO □□□ □□□□ □□ □□□□ □□ □□□ □□ □□□ □□:

□□	□	□□□
Flat-R □□ (□□□ 66 □□□ □□ □ □)	27.4%	< 30% □□
R-range □□□ (max − min)	0.200	> 0.02 (10× margin)
F1-Faithfulness □□	0.22	□□ □□

□ 5,000 □□ □ □ 72.6%□□ □□□ □□□ □□ □□□ □□□□□ □□ □□, PPO □□ □□□□ □□□□. □□□ □□

□□ scripts/cache_quality_report.py.

2.3 Reward □□□ □□ □□

EM = 0 □ □□ □□ □□□□□ □□□□□, □□ reward □

R ≈ 0.5·F1 + 0.2·Faithfulness − penalty

□□. □ □□□□ F1 □□□ 0.5 □ Faith □□□ 0.2 □ □□ □□ 5:2 □ □□□□. □□□ 0.3 □ EM □□ 0 □□ □

□□□ □□□ □□□□ □□□. □□□ □ □□□ □□□□□ □□□□ □□□□ (§8 □□ □□ □□□ □□ □□).

3 □ □□ □□ □□ (5,000 QA)

3.1 [□ 6-2] □□ □□ □□ (□□ □□□)

□□	**F1_strict**	**F1_substring**	**Faithfulness**	**Total R**
Vector-only	0.082 ± 0.17	0.334 ± 0.41	0.742 ± 0.43	0.187 ± 0.15
Graph-only	0.043 ± 0.14	0.196 ± 0.38	0.661 ± 0.47	0.080
Ontology-only	0.045 ± 0.15	0.220 ± 0.38	0.711 ± 0.45	0.164 ± 0.13
Uniform (1/3, 1/3, 1/3)	0.083 ± 0.17	0.362 ± 0.42	0.844 ± 0.35	0.209 ± 0.12

R-DWA (□□ □□ □□□□□)	0.137 ± 0.19	0.450 ± 0.44	0.835 ± 0.35	0.199 ± 0.12
L-DWA (□ □□, 3 seeds)	0.167 ± 0.001	0.488 ± 0.006	0.866 ± 0.004	0.215 ± 0.002
Oracle (per-query argmax)	0.168 ± 0.19	0.483 ± 0.43	0.888 ± 0.30	0.250 ± 0.11

3.2 □□ □□

11. L-DWA vs R-DWA: F1_strict +21.9% (0.137 → 0.167), F1_substring +8.4% (0.450 → 0.488), Faithfulness +3.6% (0.835 → 0.866), Total R +7.6% (0.199 → 0.215).
12. L-DWA □ Oracle □□□ □□:

- F1_strict: 0.167 / 0.168 = 99.4% □□
- Faithfulness: 0.866 / 0.888 = 97.5% □□
- F1_substring □□□ Oracle □□ (0.488 > 0.483) → □□ □□□ □□ □ □□□□ robust □□□

13. 3 seeds □□□: std □□ 0.006 □□. □□ □□□□□ □□ □□ □□.

▸ [□□ 6-5] Oracle □□ — □□□□ □□

>

▸ □□. □ □□□ □□ □□ □□□ □□ □□ □□ total_reward □ □□□□□ □□□ □□□ □□□□ □□□□ □□.

>

▸ □□.

▸ π_{oracle}(q) = argmax_{a∈A_{grid}} R(q, a), A_{grid} = Δ³ □ 10-grid □□□ (66 □ □)

>

▸ □□. Gold □ □□□ R □□ □□ → □□ □□ □□. □ □□□□□ □□ □□□ □□(ceiling) □□□□ □□ □□ □□ □□ □□.

>

▸ □□. L-DWA □ Oracle □ □□□□ □□ (Oracle gap recovery) □ □□ □□□□ □□.

>

▸ | □□ | R-DWA | L-DWA | Oracle | Gap recovery |

▸ |---|---|---|---|---|

▸ | F1_{strict} | 0.137 | 0.167 | 0.168 | 96% |

▸ | Faithfulness | 0.835 | 0.866 | 0.888 | 57% |

>

▸ L-DWA □ Oracle □ 96% □□ → "□ □□□□□□ □ □□ □□□ □□□ □□□ □□ □□". □□ 4% □ □□ □□ □□ (□□ 5-1 State 18-dim) □□ RL-specific noise □ □□.

3.3 □□ JKSCI □□ □□ □□

□□	□□ □□	**Sentence** □□□ □ □□	**List** □□□□ □□	□□□
R-DWA F1	0.86	0.137 (strict) / 0.450 (substring)	0.529 (strict) / 0.482 (substring)	strict 16% → 61%, substring 52%
R-DWA EM	0.78	0.000	0.387	□□□ 0 → 50%
R-DWA Faith	0.89	0.835	0.544	94% / 61% (□□ □□ □)
Simple-only F1_{strict}	—	0.153	0.874	JKSCI 0.86 □□ □ □

□□. Sentence □□□□ □□□□□□ F1 gap □ □□□ □□ □□□ □□ □□ □□ □□□ □□□□ □□□ (List □□□ □) F1_{strict} □ R-DWA □□ 0.529, □□ Simple □□□ □□ 0.874 □ JKSCI □ 0.86 □ □□□ □□□□. □□ □□ □□□ Simple-subset □□ short-list gold □□□□□□ □□□□, □ □□□ □□□ □□(§3.4)□□ □□□□ □□□ □□□ □□□□.

Faithfulness □ sentence-vs-list □□(0.835 → 0.544)□ 2-branch □□(□□ 6-4) □ □□□□. □□□ □□□ □ □□ □□□ □□ claim □□ □□□□□ □□□ □□□ □ □□□ □□□□. □□ □□ □□□ □□ □□□ □□□ □□, □□□ □□□ CS-7.2 □□ □□□□□ □□□□.

3.4 □□□□ □□□ □□ □□□ (List Prompt □□)

□ □□□ PROMPT_TEMPLATE_LIST □ □□□□ □□ pipeline · □□ gold · □□ retrieval □□□ □□ □□ □□□ □ □□ □□□ □ A/B □□□ □□□□□. 5,000 QA □□ □□ □□(□ 6-2b) □ □□(□□ 6-1):

[□ 6-2b] List □□□□ □□ □□ (5,000 QA)

□□	**F1_{strict}**	**F1_{substring}**	**F1_{char}**	**EM**	**Faithfulness**
R-DWA	0.529 ± 0.43	0.482 ± 0.44	0.469 ± 0.44	0.387	0.544 ± 0.49
Oracle	0.554 ±	0.504 ±	0.487 ±	0.388	0.570 ±

	0.42	0.44	0.43		0.49
L-DWA (3 seeds)	0.562 ± 0.007	0.507 ± 0.009	0.494 ± 0.010	0.388 ± 0.001	0.580 ± 0.009

- L-DWA > Oracle in list prompt — □□ Dirichlet □□□ 66-□ □□ □□□ □□□□.
- Simple □□ L-DWA F1_{strict} = 0.906 (□□ 6-5 per-type □).
- Conditional □□ L-DWA F1_{strict} = 0.304 (R-DWA 0.223 □□ +36.7%) — □□□ □ □□□ □□ per-type claim.
- □□□ □□/□□□ □□ results/m8_list_prompt_summary.md □ □□ □□□.

3.5 Faithfulness □□ □□□

□ □□□ Faithfulness □ □ □□ □ □□□□ (□□ 6-4). □□□ □□□ □□ Faithfulness □□□

"prediction □ □ □□□□" □ □□□ "□□□ □ □□□□□" □ □□□□. □□:

14. CS-7.2: sentence □□□ 5 □ □□□ □□□□ filler □□□ any-token □□□ □□□□ Faith 1.0 □ □□. □□□ 1/4 □ □□ + 1 □ □□ + "□□ □□" □□.
15. List □□□ □□ □□/□□ □□□□□ per-item □□□ Faith 0.5 (1/2 □□ □□) □ □□ □□□ □□□ □□.

□□ −33% □□□ □□□ □□□ □□ □□ □□ □ □□ L-DWA □ □□ □□ □ □□□ □□ □□□ □□□ □□ (□□ RAGAS LLM-as-judge □ □□ □□ □□ — future work).

□□□ □□□ □□ (□□ □ future work). □□ □□□ hybrid-rag-comparsion □ □ □□□ □□ □□ □□ + List □□□□ □□□ □□□□ □ □□□□ □□ □□ □□ □ □□□□. □□ □□ □□ □□□□□ □□□ □□ □□□ □□□□.

4 □ □□ □□□ □□ □□

4.1 [□ 6-3] □□□ □□ (□□ □□□ □□ □□ — ⚠)

> ▸ □□: □ §□ □□ □□-□□ □□□□ per-type □□□□ □□□□ □□□. □□ □□ □□ □□ F1_strict □□, F1_substring □ §3 □ □□□□. □□ 3 □ □□ per-type □□□□ □□ \$6 □ □□ □□□□ □□ follow-up □□ □□□.

Simple (n=2,000):

□□	**F1_strict (□□)**	**F1_substring**
R-DWA	0.153	0.837
L-DWA	0.174 (+13.6%)	0.861 (+2.9%)
Oracle	0.189	0.859

Multi-hop (n=1,750):

□□	**F1_strict (□□)**	**F1_substring**

R-DWA	0.017	0.261
L-DWA	0.019 (+13.1%)	0.261 (+0.04%)
Oracle	0.030	0.264

Conditional (n=1,250) — □ □□□ □□ □□ □□:

□□	**F1_strict (□□)**	**F1_substring**
R-DWA	0.019	0.090
L-DWA	0.042 (+119.2%)	0.208 (+131.4%)
Oracle	0.044	0.195

4.2 Conditional □□ □□

□□□ □□ ("40 □ □□ □□□ □□ □ □□□ □□□□?") □ □□·□□·□□ □□□ □□□ □□ □□□ □□□. R-DWA □ Table 4-1 □□ □□□ (α=0.2, β=0.2, γ=0.6) □ ontology □ □□ □□□□ □□□□□, □□ F1_substring □ 0.090 □ □□□. R-DWA □ □□□ □□□ □□ □□ (□□ □□ □□, □□□ □□ □□ □) □ □□ □□ □□□.

L-DWA □ □□ □□□□ F1_substring 0.208 □ □□ □□, □□:

- R-DWA □□ +131% □□ □□
- Oracle □ 0.195 □ □□ — Oracle □ □ □□□ □□ cache □□ □□ □□□□ □□□□□, strict F1 □ □□□ □□□ Oracle □ substring □□□□□ sub-optimal □ □ □□. L-DWA □ strict □□□□ □□ □□□□□ substring □□□ robust □ □□□ □□□□□.

□□ per-query □□□ □□□□ □□ □□□ □ □□□□ □ □□□ □□ □□□ □□□□.

4.3 Multi-hop □ □□□ □□

Multi-hop □□ F1_substring □□□ +0.04% □ □□□ □□□□. R-DWA □ □□ β=0.6 □□ Graph □□□□ □□ □□ □□, Graph BFS □□ □□□ □□ □□□ □□□□ □□□□. L-DWA □ Faithfulness □□ +3.5%□ □□□□ □□□ □□ □□ □□ □□□□ □□□□□□ □□□□.

5 □ PPO □□ □□□

5.1 3-seed □□

□ seed □ early R (□ 100 ep) □ late R (□□□ 100 ep):

Seed	**early R**	**late R**	**Δ**	□□

42	0.1988	0.2145	+0.0157	□ learning
123	0.1970	0.2153	+0.0183	□ learning
999	0.1964	0.2146	+0.0182	□ learning

□□ +0.016~0.018 □ □□□ reward □□ — □□ □□□□□ □□ □□□ □□.

5.2 □□□ F1 □□□ □□ □□□

□□ □□ □□ □□□□ □□□ □□□ □□□□ □□□□, □ □□□ L-DWA □ □□ □□□□ □□□□ □□ □□□ □□□□□:

16. □□□ F1 □□ □□ (0.072→0.137 □□ □) □□□□, □□□ □□□□ □□□.
17. □ □□□ □□ A □ B □□ □□ □□□ □□ □□ □ □□ □□ □□□□ □□□□.
18. PPO □ □□ □□ □ □□ □□ □□ □ □□□ □□□□.
19. □□□ 3 seed □□ □□ □□□ (R ≈ 0.215) □□□ □□ □□ □□□□.
20. □□□ □□ □□□ □□ □□□□ □□ (F1_strict 0.167) □□ □□□ □□.

□ □□□ buggy F1 □ □□□□□□□ □□□ □□□ □□ F1 □ □□ □□□. □ □□□ □□□ □□□□ □□ □□ □□□□□.

5.3 □□ □□ □□□ □□ (seed 42 □□)

L-DWA □ □□ □□ □□□ ≈ (α=0.25, β=0.30, γ=0.45). R-DWA □ (0.25, 0.45, 0.30) □□ β → γ □□□. Oracle □ □□□ γ-□□ (0.11, 0.16, 0.73) □ □□ □□□□□ □□□ □□ □□ □□ □□□□. □ □ entropy □□ 0.01 □ □□□ □□□□ Dirichlet □□□ smoothing □□□ □□□ regularized □□ □ □□□ □□□□.

6 □ □□ □□

□□	□
PPO □□□□ □□□□ □	5,636
□□ □□ (seed 1 □, RTX 4090)	16 □ (training) + 20 □ (state pre-compute □ seed □)
□□ API □□ □	0 (offline cache)
Inference latency	< 3 ms (FAISS local + MLP forward)
□□ M5 + M6 □□	≈ \$38 (\$33 □□ + \$5 □□)

□□ □□ R-DWA □□ □ □□□ inference □□□□□ □□ □□□□, □□ 1 □□ □□□ □□ RL □□ RAG □□□ \$100+ □□□ □□ □□□□.

7 □ □□ □□□ □□

7.1 □□

□□ □□□□ □□□ L-DWA □ HotpotQA/MuSiQue/PubMedQA □ 3 □ □□ □□□□□ naive transfer □□□. □ □□□□□ □□□ context paragraphs □ □□ □□□□□ per-query FAISS □ □□□ □ Vector □□□ □□. Graph / Ontology □ □ □□□□ □□□ □□□□ □□ □□□□ □□□ □□ — α □□□□ □□□□ □□□□.

7.2 [□ 6-8] □□ □□□ □□

HotpotQA Hard 300 (□□ multi-hop)

□□	**F1_strict**	**F1_substring**	**Faithfulness**
R-DWA	0.096	0.357	0.720
Vector-only	0.101	0.378	0.777
L-DWA (univ-trained)	0.074 ⬇	0.249 ⬇	0.600 ⬇

MuSiQue Dev 300 (4-hop □□)

□□	**F1_strict**	**F1_substring**	**Faithfulness**
R-DWA	0.046	0.123	0.415
Vector-only	0.056	0.145	0.480
L-DWA (univ-trained)	0.038 ⬇	0.099 ⬇	0.358 ⬇

PubMedQA Pharma 300 (biomedical yes/no + long answer)

□□	**F1_strict**	**F1_substring**	**Faithfulness**
R-DWA	0.214	0.004	0.757
Vector-only	0.231	0.006	0.812
L-DWA (univ-trained)	0.149 ⬇	0.005	0.546 ⬇

7.3 □□ □□: 3-□ □□ □□

□ □□□□ □□□□ □□ □□ L-DWA □ R-DWA □ Vector-only □□ 30~35% □□ □□. □ □□□ □ □□ □ 3 □□ □□ □□□□ □□□□.

□□	□□	□□ □□□
L1 □□ □□	□□□ intent regex □ □□ □□□ □□ density=(0,0,0) □□	□ intent □ □□ □□ □□ □ □□
L2 □□□ □□	□□ □□ (□□ □□□) □ □□ □□ (□□ □□·□□ □) □ mismatch	⚠ □□ □□ (□□□ □□ □□□)
L3 □□□□ □□	HotpotQA/MuSiQue □ Graph·Ontology □□ □□ —	□ □□ □□ □□□□□□□ □□ □□

	Triple-Hybrid □□ □□ □□	

□ □□ §7.4 ~ §7.7 □ □ □□□ □□ □□ □□ □□.

7.4 HotpotQA □□□-□□ □□□ □□

L3 □□ (Graph/Ontology □□) □□: HotpotQA □□□ L-DWA □ □□□ (cache 20K □□□ \$6, □□ 16 □).

- □□ trajectory FLAT: R_early = 0.203, R_late = 0.202, Δ = −0.001
- □□ □□: F1_strict 0.101 — Vector-only 0.102 □ □□□ □□.

□□: HotpotQA □ Graph/Ontology □ □□□□ merge_contexts □ α □ □□□□□ □□. □□ □□□ □□□□ α ≥ 0.2 □ □□ □□□ top-3 contexts □ □□ □□ → □□ □□ plateau → PPO gradient □ □.

□□: L3 □ □□□ □□ □□□□□□ □□ □□. Triple-Hybrid □ Graph/Ontology □ □□□ □□□□□□ □□.

7.5 Option A — □□ intent □□ □□ □ □□□

L1 □□ (□□ □□) □□ □□. RuleBasedIntent □ □□ entity/relation/constraint □□□ □□□□ (□ □□ □□□ □□) HotpotQA 300 QA □ □□□.

□□	**F1_{strict} (Korean-only intent)**	**F1_{strict} (English-aware intent)**	**Δ**
R-DWA	0.096	0.132	+38%
L-DWA (univ-trained)	0.074	0.110	+49%

□□:

- Density □ □□ □□□□ (0,0,0) → (0.8, 0.25, 0.33) □ □□□□ □□
- R-DWA, L-DWA □□ □□□ □□ — L1 □□ □□
- L-DWA □□□□ □ □ (+49% > +38%) — state-□□ □□□ density □□□□□ □ □ □□
- □□□ L-DWA < R-DWA (0.110 < 0.132) — L2, L3 □□

7.6 Option B — □□ □□ □□ □□□□ □□

L2 □□ (□□□ □□) □□: □ □□ □□□ □□ □□□□ □□□ □□ (data/university_en/).

□□□ □□: 50 depts · 100 profs · 200 courses · 100 projects · 414 docs · 403 gold QA (Simple 200 / Multi-hop 114 / Conditional 89). □□ □□□□:

data/university_en/dataset_generator_en.py.

R-DWA □□ □□ □□ (list prompt, Vector-only □□ — Graph/Ontology □□□ □□□):

□□	**Korean □□ (5,000 QA)**	**English □□ (403 QA)**	**Δ**
F1_{strict}	0.529	0.663	+25%
F1_{substring}	0.482	0.609	+26%
F1_{char}	0.469	0.591	+26%
EM_{norm}	0.387	0.288	−26%
Faithfulness	0.544	0.958	+76%

□□ (□□):

21. R-DWA □ □□ □□□□□□ F1_{strict} 0.663 — Korean □□□ 0.529 □□ 25% □□. □□□□ □□□□ □□□ □ □ □□.
22. Faithfulness 0.958 — □□ □□ □□. □□ □□□□ 1-to-1 □□ + □□ □□□ □□□ per-item □□ □□.
23. EM □ Korean □□ □□ □ gold □□ □□ ("N years old" □ □□□) □□ □□ □□ □□.

□□: L2 □□ □□ — □ □□□ □□□□ · □□□□□ · □□□□ □□ □□□. □□ □□ □□□□ □□□□ R-DWA □ Korean □□ □□□ □ □□.

7.7 Option C — □□ □□ L-DWA □□ □ □□

Option B □□□ □□ L-DWA □ end-to-end □□□□ R-DWA □□ □□ □□ □□□ □□□□□.

□□ □□:

- Cache □□: 403 QA × 66 gt = 26,598 entries, □ 44 □, \$5 (scripts/build_cache_en.py)
- PPO □□: seed 42, 10,000 episodes, GPU (RTX 4090), □ 16 □ (scripts/train_ppo_en.py)

- State: □□ □□□ RuleBasedIntent □ □□□ regex □□ — □□ □□□ □□ density=(0,0,0). □ Option A □ □□ intent □□□ □□□ □□ □□ (□□ □□ □□).

□□ (403 □□ QA):

□□	**R-DWA**	**L-DWA (EN-trained, Korean intent)**	**Δ**
F1_{strict}	0.663	0.661	−0.2%
F1_{substring}	0.609	0.607	−0.3%
F1_{char}	0.591	0.588	−0.5%
EM_{norm}	0.288	0.283	−1.7%
Faithfulness	0.958	0.955	−0.3%

□□ (□□□ □□ □□): L-DWA ≈ R-DWA. PPO □□□□ □□□ □□ □□□ □□□□ □□.

□□ □□:

- State □ □□ □□ □□□ □□ (0, 0, 0) + □□ query_meta → 18-dim state □ □□ (near-constant).
- Dirichlet policy □ □□ □□□ □□ □□ □□ → □□□□□ "□□□ □□ □□□" □□□□ □□.
- □□ PPO □□ □□ R-DWA □□ □□□ □□□ □□. □□□ L-DWA ≈ R-DWA.

HotpotQA □ □□□ EN-trained L-DWA:

- F1_{strict} 0.149 (univ-trained L-DWA 0.110 □□ +35%)
- □□□ R-DWA 0.132 □□□ □□ □ — □□ Oracle □ □□□ "□□ □□ □□□" □ □□□ □.

7.8 □□ — □□ □□□ □□□ □□ □□

□□	□□	□□□
Korean-trained L-DWA → HotpotQA	F1 −30~35%	L1+L2+L3 □□ □□
Option A: EN intent □□	R-DWA +38%, L-DWA +49%	L1 □□ □□
HotpotQA □□ □□□ (L3 test)	□□ FLAT, Vector-only □□	L3 □□ □□ (Graph/Onto □□)
Option B: EN □□ □□□□ R-DWA	F1 0.663 (Korean □□ +25%)	L2 □□ — □□□ □□ □□
Option C: EN L-DWA □□	L-DWA ≈ R-DWA (Korean intent)	State □□□ L-DWA □□□ □□

□□ □□ (□□ □□□ □□□ □□□□□):

24. L1 (□□): □□ □□ □□ — □□□ intent □□□ (□□ □□).
25. L2 (□□□ □□): □ □□ □□□ □□□ □□□□ □□□ □ □ □□ — □□ □□.
26. L3 (Graph/Ontology □□): Triple-Hybrid □ □□□ □□ □□ — □□□ □□□ □□ □□□□□□□□ L-DWA □ □□ □□.
27. L-DWA □□□ □□ = state □□□: Option C □ □□ state □□ → □□ □□ state □□□ cross-lingual □□□ □□.

□ □□□ □□□ "Triple-Hybrid □□□ □□□ □□□ □□□□□ L-DWA □ Oracle □ □□" □ □□□□ □□ □□ □□, □□·□□□ □□□ □ □□ □□ □□ □□□ □□□ □□ □□□ □□□ future work □□ (§8.2).

8 □ □□ □ □□ □□

8.1 □ □□□ □□

28. □□□ □□□: F1_strict □ F1_substring □ □□ □□□ □□ 3~4 □ □□□ □□ □□□, □□□ □□□ □□□ □□ □□□ □□ □□□ □□□□ □□ □□□□ □□□□ □□□□.
29. □□ □□ □□□: □□ □□ □□□□□□□□ end-to-end □□□ □□□□□. §7 □ □□ □□□□ transfer feasibility □ □□.
30. EM = 0 □□□: □□ □□□ gold □□□□ EM □ □□ □□□□ 0 □□. Reward □□ 0.3·EM □□ □□ □ □□ □□□□ □□□□. □□□ □□□ (□: 0.7·F1 + 0.3·Faith) □ □□□□ □□ □□ □□□□.
31. BERT intent classifier □□□: src/intent/bert_classifier.py □ klue/bert-base □□ □ □□□ □□□ □□□□ □□□ □□□ □□ □□. intent_logits=(0,0,0) □□□□□ L-DWA □ Oracle 99.4% □□□□ □□□ □, BERT □□□ □□ □□□ □□□ conditional □□□□□ □□ □□ □□□ □□.
32. Cache □□ □ □□□ □□: §1.4 □ □□□ □□□ □□□ □□ cache □□ □ F1 □□□ □□□□□□, §5.2 □□ □□□□□ □□ □□ □□ □□ □□ □□□□ □□□□.

8.2 □□ □□ □□

33. □□□ state feature: multilingual BERT □□ intent encoder □□ retrieval □□□□□ □□ □ domain-invariant state.
34. □□□□ fine-tuning: □□ per-domain cache (□: 500 QA × 66 = 33K □□□, ~$3) □□ □ episode □□ □□.
35. Reward □□□: EM □□□ 0.3 □ F1 (list-aware) □□ Faithfulness □ □□□.
36. Joint training: □□ □□□□ cache □ □□ □□□□ L-DWA □ □□□ □□□ signature □ □□□□□ □□.

37. □□ □□ □□: □□ hybrid-rag-comparsion □□□□ □ □□□ □□ □□ (□□□ □□) □ □□□□ F1 □□ □ □ □□□□□□□ □□.

9 □ □□

□ □□ □□□ □□□□□.

- L-DWA □ R-DWA □□ F1_strict +21.9%, F1_substring +8.4%, Faithfulness +3.6% □ □□ □□□ □□.
- L-DWA □ per-query argmax Oracle □ 99.4% □□ □ □□□□, F1_substring □□□ Oracle □□.
- Conditional □□□□ F1_substring +131% — □ □□□ □□ □□□ per-type □□.
- 3-seed □□□□ 0.006 □□ □ □□ □□□ □□.
- □□□□ □□□ □□ □□ 85% □□, □□ 98% □□ ($288 → $33, 50h → 1h PPO).
- □□ □□□ □□ □□ naive transfer □ □□□ □□□ □□, □□□ □□ □□□ □□□□ □□.

□□□ □□□ Ch.7 □□□□ □□□□.

CS-1 ~ CS-6. □□□ □□□ □□

▸ □□. Ch.VI § 8 □□ □□ □ □□. List-prompt □□□ CS-7 (□□ □□) □ □□ □□.

▸ □□ □□. □□ □□ · □□ retrieval · □□ □□ □□ □□ □□□□ □□ □□□ □□□, □□ □□□ □ □□□ □□ □□ □ □□□□□□□ □□.

CS-1. Simple □□□□ L-DWA □□ (□□)

▸ □□. "□□□ □□□ □□ □□□?"

▸ Gold. □□□□□□□

▸ □□. Simple (2,000 □ □ □□□)

□□	□□□ **(α, β, γ)**	□□	**F1_{strict}**	**EM**
R-DWA	(0.2, 0.6, 0.2)	□□□□□□□	1.00	1.00
L-DWA (seed 42)	(0.5, 0.4, 0.1)	□□□□□□□	1.00	1.00
Oracle	(0.7, 0.1, 0.2)	□□□□□□□	1.00	1.00

□□. Simple □□□□ □ □□ □□ □□ □□. □□□ □□□ □□ □□ (□□ 6-3)□ □□:

- R-DWA □ query_type □□□ □□ (0.5, 0.3, 0.2) □ □□□ 6 □□ □□.
- L-DWA □ □□ Dirichlet □□ → □□ □□ □□.
- Oracle □ 66-□ □□ □□ □ argmax → Ontology-heavy □□.

□□ □□□. L-DWA □ Oracle □ 96% □ □□□□ □□□□□ Simple □□□□ "□□ □□□ □□ 1.0 □ □□ □ □" □ □□ □□ (simple subset F1_{strict}: R-DWA 0.874, L-DWA 0.906, Oracle 0.901).

CS-2. Multi-hop □□ □□

- ▸ □□. "□□ □□□ □□□□□□□□ □□ □□□□□ □□□ □□□?"
- ▸ Gold. □□□□□□□□□□, □□□□
- ▸ □□. Multi-hop (1,750 □ □ □□ □□□ □□)

□□	□□	□□□ **gold** □□	**F1_{strict}**	**F1_{sub}**
R-DWA	□□□□□□□□□□	1/2	0.67	0.50
L-DWA	□□□□□□□□□□	1/2	0.67	0.50
Oracle	□□□□□□□□□□, □□□□	2/2	1.00	1.00

□□. L-DWA □ R-DWA □ □□□ □□-□□□ □□□ □□. Oracle □□ □ □□□□ □□ □□. □□ multi-hop □□ (n=1,750) □□ L-DWA F1_{strict} 0.365 □ R-DWA 0.354 □□ +3.1% □ □□.

□□ □□□. Multi-hop □□□ □ □□□ □□ □□. Ch.Ⅶ □□□□ "□□□ hop □□□ □□ □□ □□ (18-dim) □ □ □□□ □□□□□ □□□, □□ multi-turn state □□ dedicated graph-path encoder □ □□ □□" □ □□.

CS-3. Conditional □□ □□

- ▸ □□. "□□□□□ □□ 55 □ □□ □□□?"
- ▸ Gold. □□□, □□□, □□□, □□□, □□□, □□□, □□□, □□□, □□□ (9 □)
- ▸ □□. Conditional (1,250 □ □ L-DWA □ sweet-spot □□)

□□	□□□	□□	□□	**F1_{strict}**	**Δ R-DWA**
R-DWA	(0.2, 0.2, 0.6)	□□□ (□□)	0/9	0.00	□□
L-DWA	(0.5, 0.4, 0.1)	□□□ (□□)	0/9	0.00	0%
Oracle	(0.1, 0.1, 0.8)	□□□, □□□, …, □□□ (9/9)	9/9	1.00	+∞

□□. □ □□ □□ ontology □□ "55 □ □□" □ □□□□ □□ □□ □□. □□□ □□ conditional □□ (n=1,250) □□□:

□□	**F1_{strict} (conditional)**	**Δ R-DWA**
R-DWA	0.223	□□
L-DWA	0.304	+36.7%
Oracle	0.290	+30.0%

□□ □□□. Conditional □□□□ L-DWA □ Oracle □□ (0.304 > 0.290). □□ □□ 6-2 □□□□ □□ □□□□, □ □□□ □□ □□ per-type claim. L-DWA □ □□□ Dirichlet □□□ 66-□ □□ □□□ □□□□ per-query □□□□ □□□□□ □□□□ □□.

CS-4. □□ □□ □□ (Faith 2-branch □□)

- ▸ □□. "□□□□□□□□ □□ 55 □ □□ □□□?"
- ▸ Gold. □□□, □□□, □□□, □□□
- ▸ □□. Conditional, □□ □□

□□□□	□□	□□ **per-item**	□□□ **Faith (any-token)**	□□□□ **Faith (per-item)**
Sentence	(5 □ □□□ □□)	1/2 □□ □□	1.00 (inflated)	—
List	□□□, □□□	1/2 (□□□ □□)	—	0.50 (□□)

□□. □ □□□□ □□ □□ 4 □ □ 1 □ (□□□) □ □□ + □□□ □□. Sentence □□□□□ any-token □□□ "□□", "□□" □ filler □□□ □□ context □□ □□□□ Faith 1.0 □□ □□□□. List □□□□□ per-item □□□ □ □□□ context □ □□ □□□□□ □□ □□ → Faith 0.5 □ □□ □□ □□ □□.

□□ □□□. □ □□□ Faithfulness 2-branch □□ (□□ 6-4, □□ C.2) □ □□ □□ □□□ □□□ □□ □□□ □□ □□ □□. □□ □□ □□ □□□ □□ □□ (□□ C.1) □ □□ □ □□□ □□ □□□ □□.

CS-5. Oracle □ □□□ hard case

- ▸ □□. "□□□□ □□ 40 □ □□ □□□ □□ □□□?"
- ▸ Gold. □□□□□□□□□, □□□□, □□□□, … (12 □ □□□ □□ □□)
- ▸ □□. Conditional + □□□ □□□□□

□□	□□□	□□	**F1_{strict}**
R-DWA	(0.2, 0.2, 0.6)	□□□ □□ □ □□□□	0.00
L-DWA	(0.5, 0.4, 0.1)	□□□ □□ □ □□□□	0.00

Oracle	(0.3, 0.3, 0.4)	□□□□□□□□□, □□□□ (2/12)	0.25

□□. □□□□ □□ □□□ retrieval □□□ □□□□□□ □□□□, □ □□ □ Oracle □□ □□□□ □□. R-DWA □ L-DWA □ "□□ □ □□" □□ □□.

□□ □□□.

38. Retrieval □□ □ DWA □□□ □□ □□ — □□ □□□ context □ □□ □□□ □□ □□. Oracle □□ 2/12 □ □□.
39. L-DWA □ □□ 4% gap (Oracle □ 96% □ □□) □ □□□ retrieval-bound □□□□ □□. Ch.VII □□□□ "retrieval □□□ future work □□□□" □ □□.

CS-6. Cross-domain □□ □□ (HotpotQA)

▸ □□ (□□). "Which magazine was started first, Arthur's Magazine or First for Women?"

▸ Gold. Arthur's Magazine

▸ □□. HotpotQA Hard comparison (□□ 2-hop)

□□	**density □□**	□□□	□□	**F1_{strict}**
Vector-only	—	(1, 0, 0)	Arthur's Magazine	1.00
R-DWA (univ-trained rules)	(0, 0, 0) ⚠	(0.33, 0.33, 0.33) fallback	(□□ □□)	0.25
L-DWA (univ-trained)	(0, 0, 0) ⚠	(0.35, 0.30, 0.35)	(□□ □□)	0.30

□□. □□□ regex □□ intent □□□□ □□ □□□□ □□ density □ 0 □□ □□. □ out-of-distribution □□□□ L-DWA □ □□□ policy □ □□□□□ □□ □□ → Vector-only baseline □□□□ □□□□□ □□□ □□ □□.

□□ □□□.

40. Cross-domain □□□ naive □□ □□. □□□ □□ □□□ □□ □□□ □□□ □□□ □□□ □□.
41. Ch.VII future work: language-neutral intent encoder (multilingual BERT, □: XLM-R) □ □□□ □□.
42. □ □□□ □ □□ □□ □ □□□ □□ □□ □□□□ □□ □□□□□□, □□□ L-DWA □ Oracle □ 96% □ □□ + F1_{substring} □□ Oracle □□ — □□□ □□ □□ □□□ □ □□□ □□.

□□ — 6 □□□□ □□□ □□ □□

CS	□□	L-DWA □□	□ □□□ □□ □□?
1	Simple	R-DWA □ □□ (□ □ F1=1.0)	Simple □□ □□ □□□ □ □□ → □□ □□ □□□ □□
2	Multi-hop	R-DWA □ □□ (□□ □□)	Multi-hop □ □□ □□ → future work (multi-turn state)
3	Conditional	R-DWA □□ +36.7% □□	□□ per-type claim, Oracle □□
4	Conditional + □□	□□ □□ □□	□□ □□□ □□ (2-branch Faith)
5	Hard conditional	Oracle □ □□	Retrieval □□ □□ → future work
6	Cross-domain	Vector-only □□ □□	□□□ □□□ feature, □□ □□ □□□□ future work

□ 6 □ □□□ + CS-7 (list prompt □□) □ L-DWA □ □□□ □□, □□ □□, □□ □□□ □□ □ □□ □□□□.

CS-7. List-style □□□□ □□ □□□ □□□

▸ □□. Ch.VI § 3.4 □□□□ □□□ □□ □□□ □ □□ □□.

▸ □□. □□ □□·□□ □□□·□□ LLM □ □□ sentence □□□□□ list □□□□□ □□ □□·□□ □□ □□□ 7 □ □□□ □□□□. □□ □□□□□ F1_{strict} □ □ 3~10 □ □□□□ □□□ □□, Faithfulness □ 2-branch □□(□□ 6-4) □ □□□ □□□ □□□□□ □□□□.

CS-7.1 □□ 1: □□ □□ □□ □□ (Simple)

▸ □□. "□□□□ □□□□□ □□□ □□□□ □□□?"

▸ Gold. □□□, □□□, □□□

▸ R-DWA □□ □□□. (α, β, γ) = (0.2, 0.6, 0.2)

Sentence □□□□ □□

□□□□ □□□□□ □□□ □□□□ □□□ □□□, □□□, □□□□□□.

List □□□□ □□

□□□, □□□, □□□

□□ □□

□□	Sentence	List
F1_{strict}	0.364	1.000
F1_{substring}	1.000	1.000
EM_{norm}	0.000	1.000
Faithfulness	1.000	1.000

Latency (s)	2.08	0.79

□□. □□□ □□□□ sentence □□□□□ "□□□", "□□□□", "□□□", "□□□" □□ filler □□ □□□ F1_{strict} □□ recall=1.0·precision=0.375 → F1=0.364 □ □□□□. List □ □□□□ □□ □□□ gold □ □□□ □□□□□ F1_{strict} = 1.0, EM = 1.0 □ □□□□. □□ □□ 2.6 □ □□ — □□ □□□ LLM generation time □ □□□.

CS-7.2 □□ 8: □□□ □□ □□ (Conditional, □□ □□)

- ▸ □□. "□□□□□□□□□ □□ 55 □ □□ □□□?"
- ▸ Gold. □□□, □□□, □□□, □□□
- ▸ R-DWA □□ □□□. (α, β, γ) = (0.2, 0.2, 0.6)

Sentence □□□□ □□ (□□)

- ▸ "□□□□□□□□□ □□ □□□ □□□ … □ □□□ □□ … □□□. □□□ □□□ □□□ □□□□□ □□ … □□□□ □□□ □□□ □□□□ … □□□ □□□□□□□□□ □□ 55 □ □□ □□□ □□ □□□ □□ □ □□□□. □□□□□ … □□□ □□□ □."

List □□□□ □□

□□□, □□□

□□ □□

□□	**Sentence**	**List**
F1_{strict}	0.059	0.333
F1_{substring}	0.200	0.250
Faithfulness (□□□ □□ / □□)	1.000 (□□ any-token, □□□□ □)	0.500 (1/2, □□□ □□ □□)
Latency (s)	3.68	0.74

□□. □ □□ □□ □□ 4 □ □ 1 □ (□□□) □ □□□ □□□□ □□□□. Sentence □□□□□ 5 □□ □□ □□□□ □□ □□ filler □□□ □□ □□□ faithfulness(any-token) □ □□□□ 1.0 □ □□. List □□□□□ 2 □ □□ □ 1 □□ context □ □□ □□ → Faithfulness 0.5 □ □□□ □□□ □□.

□□□. List prompt + 2-branch Faithfulness □ □□□ □□ □□ □□□ □ □□□. □ □□ 5,000 QA □□ □□□□ R-DWA □ Faithfulness □ 0.835 → 0.544 □ □□□ □□ □□ □□□ □□□ □□□ □□□. □ □□□ Ch.VI § 3.5 □ □□□□ □□.

CS-7.3 □□ 5: □□ □□ □□ □□ (Conditional, □□□ gold)

▸ □□. "□□□ □□ 40 □ □□ □□□ □□ □□□?"

▸ Gold. 14 □ □□ (□□ □□): □□□□, □□□□, □□□□□, □□□□□□□, □□□, □□□□□, □□□□, □□□□, □□□, □□□□□, □□□□□, □□□□□, □□□, □□□□

▸ List □□□□ □□. □□□□, □□□□ (2 □ □□, □ □ gold □ □□)

□□	□
F1_{strict}	0.308
F1_{substring}	0.214
F1_{char}	0.207
Faithfulness (□□□ □□)	1.000

□□. Gold list □ □□□(14 □□) □ □ LLM □ □□□□□ □□□ □□□□ □□. □□□ 2 □□□ □□ context □ □□□ → Faithfulness □□. □□□ recall □ □□□□□ F1 □ □□ □□. □□ list prompt □ □□ — □□□ □□ □□ □ □□□ □□□□, future work □ exhaustive listing prompt □□ □□□ □□.

CS-7.4 □□ 3: □□□ □□□ + gold □□ □□ (Conditional)

▸ □□. "□□ □□ □ 55 □ □□□ □□□?"

▸ Gold (□□). □ 367 □ (□□□, □□□, □□□, □□□, □□□...) — 367 □□ "5 □ + ..." □□□ □□ □ non-□□ □□.

▸ List □□□□ □□. □□□, □□□, □□□

□□	□
F1_{strict}	0.600
F1_{substring}	0.400
F1_{char}	0.560
Faithfulness (□□□ □□)	0.000

□□. Gold □ "□ 367 □" □□ □□□ list prompt □ item □□ □□□ □□□□ F1_{strict} □ □□□□□ □□□. □□ LLM □ □□□ 3 □ □□□ context □ □□ Faithfulness 0. □ □□□□□ gold □□ □□□ □ □□ □□□ □□□□ □□□□ □□□, Ch.VII □□□□ future work "□□□□ □□" □ □□□.

CS-7.5 □□ 2: □□ □□ □□ (Simple, □□ □□)

▸ □□. "□□□ □□□ □□ □□□?"

▸ Gold. □□□□□□□

□□□□	□□	**F1_{strict}**	**EM**	**Faith**
List	□□□□□□□	1.00	1.00	1.00

□□. □□ □□ □□□□ list prompt □ gold □ □□□ □□. □ benchmark □ Simple □□ (40% □□) □□ L-DWA □ F1_{strict} □ 0.906 □ □□□□ □□ □□□ □□/□□□ □□□ □□ □□□ list prompt □ sweet spot □□ □□□□.

CS-7.6 □□ 10: □□ □□ (Multi-hop, □□ □□)

- ▸ □□. "□□ □□□ □□□□□□□□ □□ □□□□□ □□□ □□□?"
- ▸ Gold. □□□□□□□□□□, □□□□
- ▸ List □□□□ □□. □□□□□□□□□□ (1/2)

□□	□
F1_{strict}	0.67
F1_{substring}	0.50

□□. Multi-hop □□□ □□□ □□□□ □□ □□□ □ □□ □□□ □□□ □□□□ □□ □□. □ □□□□ L-DWA □ F1_{strict} □ 0.365 (R-DWA 0.354 □□ +3%) □ □□ □□ □□ □□ □□□□, □ □□ □ □□ □ Ch.Ⅶ □ □□.

CS-7.7 □□ — List prompt □□ □□ □□ □□ □□

□□	**Sentence □□□□ (□□)**	**List □□□□ (□ PR)**
□□ □□	□□□ □□ (filler □□)	□□ □□ □□□ (□□□)
EM □□□	□□□ 0 (□□ □□ □□□)	0.39 (Simple □□ □□)
F1_{strict} □□	≈ 0.2 (filler □□)	≈ 0.87 (Simple □□)
Faithfulness □□□	□□ any-token (□□)	□□□ claim (□□)
□□ (sec)	1.1 ~ 3.7	0.7 ~ 0.8
□□ (□□ □□)	□□ 45	□□ 12

□□. List prompt □ □□ □□□ gold □□□ □□ □□□ □□ □□□□:

43. F1_{strict} □ R-DWA □□ +286% □□ (0.137 → 0.529)
44. EM □ 0 □□ 0.39 □ □□
45. Faithfulness □□□ □ □□ □□ □□□ □□
46. □□·□□ □□ □□ (□□ □□□)

□□ Triple-Hybrid RAG □ □□□□ □□ □□ □□□ □□□□, □ □□□ □□ □□□□ □□ □□□□ □□ □□ □□□□.

□Ⅶ□ □□ (v5 □□□)

1 □ □□ □□

□ □□ □□□ Triple-Hybrid RAG □□□□□□ □□ □□ □ □□□ □□□ Markov Decision Process □ □□□□□ Proximal Policy Optimization □□ □□□□ Learned Dynamic Weighting Algorithm (L-DWA) □ □□□□□. □□ □□ (Shin & Moon, 2025, JKSCI) □ □□ □□ R-DWA □ □ □□ □□□ □□ — (i) □□□ □□□ □□ □□□ □□□, (ii) □□ λ □□□□□□□, (iii) □□ □ □□ □□ □□, (iv) □ □□ □□ □□ — □ □□□ □□□□ □□□□, □□ □□ 5,000 QA □□□□□□ □□□ □□□ □□□□□.

2 □ □ □□□ □□

2.1 □□□□ □□

□□ 1 — MDP □□□. Triple-Hybrid RAG □ □□□ □□□ 18 □□ □□ □□□ 3-simplex □□ □□□ 1-step contextual bandit MDP □ □□□□□□. □□ □□□□□ RAG □□ □□□□□ □□ □□□ □□.

□□ 2 — PPO-based L-DWA. □□ backbone (18→64→64 Tanh) + Actor head (Softplus Dirichlet □□) + Critic head □ □□□ 5,636 □□□□□ □□ Actor-Critic □ PPO (clip ratio 0.2, 10,000 episodes) □ □□□□ 3 seeds □□ Total Reward 0.215 ± 0.002 □ □□□ □□□ □□□□□.

2.2 □□□□□ □□

□□ 3 — □□□□ □□ □□. 5,000 □□ × 66 □□ □□□ = 330,000 □□□□ reward □ SQLite □ □□ □□ □□ PPO □□ □ LLM API □□ □□ 0 □□ □□□□. 3 seeds □□□ □□□ $288 → $33 (−85%), □ wall-clock □ 50h+ → 1h PPO (−98%) □ □□□□□.

□□ 4 — □□ F1 □□ □□. □□□ □□·□□□ □□ □□□ □□□ F1_strict □ □□ □□□ gold □ □□□□ F1_substring □ □□ □□□□, □□ □□ (JKSCI 2025) □□ □□□□ □□□ □□□□ □□□□ □□□ □□□□□. □ □□□ □ □□□ □□ □□□ □□□ QA □□□ □□□□ □□ □ □□□□.

2.3 □□□ □□

□□ 5 — L-DWA □□ □□□ (5,000 QA, 3 seeds):

- F1_strict 0.167 ± 0.001 (R-DWA 0.137 □□ +21.9%)
- F1_substring 0.488 ± 0.006 (R-DWA 0.450 □□ +8.4%)

- Faithfulness 0.866 ± 0.004 (+3.6%)
- Total R 0.215 ± 0.002 (+7.6%)

□□ 6 — Oracle □□ □□. L-DWA □ per-query argmax Oracle □ F1_strict 99.4%, Faithfulness 97.5% □ □□□□, F1_substring □□□ Oracle (0.483) □ 0.488 □ □□□□. □□ □□ □□□ □□ □□ (Total R) □ □□□□ robust □□ □□□□□ □□□□.

□□ 7 — Conditional □□□ □□ □□. Conditional □□ (n=1,250) □□ L-DWA □ F1_substring 0.208 □ □□□□ R-DWA □ 0.090 □□ +131% □□ □□ □ □□□□, Oracle □ 0.195 □ □□□□. □□ per-query □□□ □□□□ □□ □□ □□ □□ □□□ □ □□ □ □□□ □□ □□□ □□□□.

□□ 8 — □□ □□□ □□□ □□ □□ + □□ (Ch.6 §7). □□ □□ L-DWA □ HotpotQA/MuSiQue/PubMedQA naive transfer □□□ □ □□□ □□□□□:

- L1 □□ □□: □□ intent □□ □□ □□□ R-DWA +38%, L-DWA +49% □□ (Option A □□).
- L2 □□□ □□: □□ □□ □□ □□□□ (403 QA) □ □ □□ □□□□ □□□ □□ R-DWA F1_{strict} 0.663 □□ — Korean □□□ 0.529 □□ +25%. □□□ □□□ □□ □ □□ □□ □□□ (Option B).
- L3 □□□□ □□: HotpotQA □□□-□□ □□□ trajectory □ FLAT (Vector-only □□) — Triple-Hybrid □ □□□ □□ □□ (Graph·Ontology □□ □□□□).
- Option C (□□ L-DWA □□): Korean-only state □□ L-DWA ≈ R-DWA — state □□□ □ L-DWA □□□ □□ □□ □□.

□ □□ □□□ □□ "naive transfer □□" □□□ "□□ □□ + □□ □□ □□" □ □□□□□. □□ □□□□ §4.2 □ □□□ intent encoder · domain-invariant state □□ future work □ □□ □□ □ □□.

3 □ □□□·□□□ □□

3.1 □□□ □□

□□□□□ RAG □□□ □□□□ □□□ □□□ □□ □□. □□ □□□ retrieval/generation □□□ self-reflection (Self-RAG) □□ classifier □□ □□□ (Adaptive-RAG) □ □□□□□. □ □□□ □□ □□ □ □ □ □□ □□□ □□ □ PPO □ □□ □□□□ RAG-RL □□□ □□□ □□□ □□□□.

□□□ QA □□□ □□□ □□. F1_strict □ F1_substring □ 0.137 vs 0.450 □□□ □□ □□□ □□ 3 □ □□ □□ □□□□ □□□□. □□ □□□ NLP □□ □□□ □□□ □□ □□□□ □□□□.

3.2 □□□ □□

□□-□□ □□ □□ □□. □□□□ □□ □□□ RL □□ RAG □□□ □□□ □□·□□□□ □□ □□ □□ (□□ $33) □□ □ □□□. □ □□□ □□ GitHub □□□ (https://github.com/sdw1621/triple-rag-phd) □ cache/university.sqlite (16.8 MB) □ 3 seed □□□□□□ □□□□ □□ □□ / □□□□ □□□□. □□□ □□ □□ □□□□ □□. □ □□□□□ L-DWA □ □□□□□ □□□□□ "~$3 □ □□ □□ + 10,000 episode PPO" □ □□□□ □□□ □□ □□□□ □□□□ (§6.7.4 □□). □□ Graph/Ontology □ □□□ □□ □□ □□□□□ (□□, □□, □□) □ □□ □□□ □□ □□□□.

4 □ □□ □ □□ □□ □□

4.1 □□□ □□

47. □□□ domain end-to-end □□. □ □□□ L-DWA □ "R-DWA □□ +21.9% + Oracle □□" □ □□□ □□ □□□□ end-to-end □□. □□ □□ □□□□ (Option B, C) □□□ R-DWA □ F1_{strict} 0.663 □ □□ □□ □□ L-DWA □ □□ □□□ state □□ □□□ □□ □□□ □□□□ □□.
48. □□ □□□ naive transfer □□. §VI.7 □□ L1+L2+L3 3-□ □□ □□ □ Option A/B/C □□ □□.
49. EM = 0 □□□ (sentence □□□□ □□). List □□□□ □□□□ □□ □□ EM 0.29, □□□ □□ EM 0.39 □□.
50. BERT intent classifier □□□. □□□ □□□□ intent_logits = (0, 0, 0) □□ □. Option C □ L-DWA ≈ R-DWA □□□ state □□ (BERT □□ □□ □□□ intent) □ L-DWA □□□ □□ □□ □ □.
51. Cache □□ □ F1 □□□ □□. §VI.1.4 □ □□□ □□ □□□ _PUNCT_RE □□□ □□, □□ □□□□ CORRIGENDUM □□ □□ □□ (2026-04-22).

4.2 □□ □□

52. □□□ state feature. multilingual BERT intent encoder □□ domain-invariant retrieval-score state.

53. □□□□ fine-tuning □□□ □□□. □□ per-domain cache + □□ PPO □ □□-□□ □□□ HotpotQA/MuSiQue/PubMedQA □□ □□.
54. Reward □□□. EM □□□ 0.3 □ F1 (list-aware) □□ Faithfulness □ □□□□ □□□ □□ □□.
55. Joint multi-domain training. □□ □□□ cache □ □□□□ L-DWA □ □□□ □□□ clustering □□.
56. □□ □□□□□ □□□ □□. hybrid-rag-comparsion □ □ □□□ □□ □□□ □□□□ F1 □□□ □ □□□□ □□ □□.
57. BERT multi-label intent □□ □ □□□. □□ Conditional □□□ □□ □□ □□.
58. □ □ Actor-Critic. □ 5,636 □□□□ □□□□□ □□□ □□□ □□□□ □□ □□.

5 □ □□ □□□ □□ □□

□ □□□ □□ □□□□ □□ □ □□□ □□-□□ □□□ □□□ □□□□.

□□, F1 □□□ □□. □□·□□□ □□□ □□□□ F1_strict □ ~50% □□ □□□□ □□ □□□ □□ □□ □□ (§VI.1.4) □□□□□. □□ □ L-DWA □ Oracle □□□□ 52% → 99.4% □ □□□ □□□□ □ □□□ □□□□ □□□ □□ □□□□□. □□□□ □□ □□□ □□□ □□□ □□□□ □□□□ □□□□.

□□, □□ □□□ transfer □□□ □□□□□. □□ □□□ "L-DWA □ 4 □ □□□□ □□□□ SOTA □ □□" □□ □ □□□□, □□ □□ □□ □□□□□. □ □□□ □ □□□ □□□ □□ "□□□ □□ □□□ □□□ □□" □ □□□□□ □□□. □□□ □□□ □□□ □□□□ □□□ □□□ □□□□ □□ □□ □□□ □□□ □□□□ □□□□.

6 □ □□□

□ □□ □□□ Triple-Hybrid RAG □ □□□ □□□ □□ □□□ □□□□ □□□□, □□ □□□ □□ □□ □□□ □□□ □ □□ □□□□ □□ □□ □□□□□□ □□ □□□ □□ □□□ □□□□□. □□□□ □□ □□□ □□□□ □□ RAG □□□ □□□ □□ □□□ □□□□, □□ F1 □□ □□□ □□□ QA □□□ □□□□ □□□□ □□□□. □□ □□□ □□□ □□□ □□ □□□ □□□□ □□ □□□ □□□ □□□□ □□.

□ □□□ □□□□ □□, □□, □□□□□□ □□ □□□ □□ □□□□ □ □□□ □□□ □□□ □□□□□, □□□ □□□□ L-DWA □ □□ □□□□□, □ □□□ □□□ □□□□ □□□ □□□□ □□□ □□□ □□□ □□□□.

□□ (Appendix)

▸ □ □□□ □□□□ □□ "□□ □□ □□□ □□ □□□ □□ □□□ □□□ □□ Triple-Hybrid RAG □□□□□□□ □□ □□□ □□"□ □□□·□□ □□□ □□ □□ □□□ □□□.

□□ A. □□□□□□□□ □□ □

A.1 PPO □□ (Ch.V Table 5-4)

□□□□	□	□□ / □□
learning rate (Adam)	3×10^{-4}	RL □□ □□
□□ □□ γ	0.99	1-step MDP □□ □□ □□
GAE λ	0.95	Schulman et al. (2015)
clip ratio ε	0.2	Schulman et al. (2017)
value coef c_V	0.5	PPO □□
entropy coef c_H	0.01	□□ □□
max grad norm	0.5	□□ □□□
total episodes	10,000	Reward □□ □□ □□
rollout per episode	32	minibatch □□ □□
update epochs	4	PPO on-policy □□ □□□
minibatch size	8	GPU □□□ □□ □□
seed	{42, 123, 999}	□□□ □□

A.2 Triple-Hybrid RAG □□

□□□□	□
LLM	gpt-4o-mini (gpt-4o-mini-2024-07-18)
LLM temperature	0.0 (□□□)
LLM top-p	1.0
LLM max tokens	500
Embedder	text-embedding-3-small (1536-dim)
Vector index	FAISS IndexFlatIP (cosine via L2 normalize)
Graph BFS max_depth	3
Ontology reasoner	Owlready2 + HermiT (□□ fallback)
Chunk size / overlap	1,000 chars / 200 chars
top_k per source	3
merge_contexts budget	9 □□ (3 × top_k)

A.3 Reward □□ (Eq. 5-7)

□□□	□
λ_{F1}	0.5
λ_{EM}	0.3
λ_{Faith}	0.2
λ_{latency}	0.1

latency threshold	5.0 sec

□□ □□. □ □□□□□□□ EM ≈ 0 □ □□□□□ □□□□ □□ Reward ≈ 0.5·F1 + 0.2·Faith − penalty □ □□□ (Ch.VI §2.3).

A.4 Offline Cache (Ch.V §4)

□□□□	□
□□□ □□□	10-grid Δ^3 (66 □ □)
□□ □	5,000
□ □□□	330,000
□□ □□	10 (ThreadPoolExecutor)
□□ □□	□ 14 □□
□□	≈ US $33
DB □□	SQLite (index on (query_id, weights))

□□ B. □□□□ □□□□

B.1 R-DWA □□ □□ □□□ □□

```
INPUT:  query_intent (type, entities, relations, constraints, density)
OUTPUT: (α, β, γ) ∈ Δ³

// 2 □□: □□ □□ → □□ □□□
if query_intent.type == "simple":
   (α₀, β₀, γ₀) = (0.50, 0.30, 0.20)
elif query_intent.type == "multi_hop":
   (α₀, β₀, γ₀) = (0.20, 0.55, 0.25)
else:  // conditional
   (α₀, β₀, γ₀) = (0.15, 0.25, 0.60)

// □□ □□□ □□ □□
(Δα, Δβ, Δγ) = density_adjustment(query_intent.density)
(α, β, γ)    = normalize(α₀ + Δα, β₀ + Δβ, γ₀ + Δγ)

return (α, β, γ)
```

□□: □□ □□ (Shin & Moon, 2025) §IV.2.

B.2 L-DWA □□ (PPO, □ □□ □□)

```
INPUT:  reward cache C (330K entries, query × weight → reward)
        initial policy θ (orthogonal init), initial value φ
HYPER:  as in Table 5-4 (lr, clip ratio, GAE λ, ...)
OUTPUT: trained Actor-Critic (θ*, φ*)

for episode in 1 .. 10_000:
```

```
    batch = []
    for _ in 1 .. rollout_per_episode:          // 32 rollouts
        q = sample_query()
        s = extract_state(q)                    // 18-dim
        c = actor_θ(s)                          // Dirichlet concentration
        a = sample_Dirichlet(c)                 // a ∈ Δ³
        a_snap = snap_to_simplex_grid(a, 10)    // 10-grid
        R = cache_lookup(C, q.id, a_snap)       // O(1)
        V = critic_φ(s)
        Â = R − V                               // 1-step advantage
        batch.append((s, a, R, Â, old_logp))

    for update_epoch in 1 .. 4:
        shuffle(batch)
        for minibatch in split(batch, 8):
            ratio = exp(logp_θ(a|s) − old_logp)
            L_CLIP = E[min(ratio · Â, clip(ratio, 1−ε, 1+ε) · Â)]
            L_V   = 0.5 · E[(R − V_φ(s))²]
            L_H   = −E[entropy(policy_θ(·|s))]
            L     = −L_CLIP + c_V · L_V + c_H · L_H
            adam_step(θ, φ, ∇L)

    log(mean_reward, entropy, policy_loss, kl)

return θ*, φ*
```

□□: □ □□ §V.3 + src/ppo/trainer.py.

B.3 L-DWA □□ (test-time)

```
INPUT:  trained Actor-Critic (θ*), query q
OUTPUT: (α, β, γ) ∈ Δ³

s = extract_state(q)            // 18-dim, □□□ □□
c = actor_θ*(s)                 // Dirichlet concentration
// mean (deterministic) □□ — □□□□ □□□□□
a_mean = c / sum(c)
return a_mean
```

□□ C. □ □□□□ □□□□ □□ □□ (□□ □ □□)

C.1 □□ □□ □□ (4/21 □□ — dff7dc1)

□□. normalize_korean □ □□·□□□·□□ □□□□ □□ □□ □□ □□□□ □□□. Gold "□□□, □□□, □□□" □ {"□□□,", "□□□,", "□□□"} □ □□□□□ prediction "□□□ □□□ □□□ …" □ {"□□□", …} □ □□ □□ □□□□ □□□ 1 □□ □□.

□□. _PUNCT_RE = re.compile(r"[,.:;!?\"'“”‘’()\[\]\{\}·□…——□]") □□ → □□□ □□□ □□.

□□.

□□	**F1_{strict} (□□ □)**	**F1_{strict} (□□ □)**	□□
R-DWA	0.072	0.137	+90%
L-DWA	0.087	0.167	+92%
Oracle	0.097	0.168	+73%

C.2 2-branch Faithfulness (4/22 □□ — □ PR)

□□. □ faithfulness() □ □□□□ □□□ □□ □ "any-token in context" □ □□ □□□ □□. □□□□ □□ (□□□ □□, □□□ □□) □ □□□ □□□ □□□□ □□□□ all-or-nothing □□□ □□.

□□. _SENTENCE_SPLIT_RE □□ □□□ □□ □ □□:

- □□□: □□ any-token (□□ □□).
- □□□□: □ □□ □□□ □□ claim □□ □□ (□□ □□□□□ □□ □□ □□ □□ □□).

□□ (5,000 QA, list prompt).

□□	**Faith (sentence □□)**	**Faith (list □□, □□)**
R-DWA	0.835	0.544
Oracle	0.888	0.570
L-DWA	0.866	0.585

Faith □ ≈ −33% □□□ □□ □□□ □□ □□ □□ □□□ □□ □□ □□ (Ch.VI §3.5, CS-7.2).

C.3 F1_{char} (□□ — □ PR)

□□. □□□ □ □□ 3-gram □□□ F1. □□·□□·□□□□ □□□ □□ □□ □□.

□□ □□. F1_{strict} □ F1_{substring} □□□ □ 3 □□. □□□ □□□□ □□ 0.494 (L-DWA 3-seed) □ □□□□ strict (0.562) □ substring (0.507) □ □□ □□.

□□ D. □□ □□ □□ (Future Work □□)

D.1 HotpotQA □□□ □□ L-DWA □□ (2026-04-21)

□□. HotpotQA Hard 300 □ □□□ □□ offline cache □ □□□□ □□ PPO □□□□ □□.

□□.

- □□ trajectory: mean_reward ≈ 0.20 → 0.20 (□□ □□, flat).

- □□ F1 0.101 (Vector-only 0.102 □□).

□□ □□.

- HotpotQA □□ □□□ Graph·Ontology □□□ □□□□ □□.
- Triple-Hybrid □ □□ □□ (3 □ □□□ □□□) □ □□□ □□ □□.
- L-DWA policy □ β ≈ γ ≈ 0 □□ □□ — "Graph/Ontology □□□ □□ □□" □□.

□□□. Triple-Hybrid □□□ □□ □□□·□□□□□ □ □□□□ □□□□□□ □□. □ □□□ Ch.Ⅶ □□□□ □□.

D.2 Cross-domain naive transfer (4-benchmark, 4/20)

□□. □□ □□ □□□□ □□□ L-DWA (seed 42) □ HotpotQA / MuSiQue / PubMedQA □ □□□ □□ □□ □□.

□□.

□□□□	**L-DWA F1**	**Vector-only F1**	**Δ**
HotpotQA Hard 300	0.074	0.101	−27%
MuSiQue Dev 300	0.038	0.056	−32%
PubMedQA Pharma 300	0.149	0.231	−36%

□□. □□□ □□ □□ intent regex □ □□ □□□□ □□ → density = (0, 0, 0). □□ state □ □□ □□ (out-of-distribution) □ □□□□ □□□ policy □ □□ □□ □□ □□.

□□□. Domain-specific □□□ □ □□□ feature □□ bug □ □□. Future work □ □□ □□ state □□ (□: multilingual BERT intent classifier) □ □□ (Ch.Ⅶ).

D.3 Sentence □□□□ □□ (v4 □□)

□□. JKSCI □ □□ □□□□ □□ (□□□ □□ □□) □□ □□.

□□. F1_{strict} 0.167 (L-DWA 3-seed). Journal □□ 0.86 □□ 19% □□.

□□. PROMPT_TEMPLATE_LIST □□□□ F1_{strict} 0.562 (JKSCI □□ 65% □ □, Simple-only □ 91% □□).

□□ E. □□□ □□ □□ □□

E.1 M1~M7 □□

□□	**Wall-clock**	□□ **(USD)**

M1 □□ □□	—	—
M2 □□ □□ □□	~8h (□□)	—
M3 BERT intent + □□ □□□	~4h (□□)	—
M4 PPO □□□	~6h (□□)	—
M5 Offline cache □□	14h (GPU)	$33
M6 PPO □□ (3 seeds, GPU)	50 min	$0 (□□ □□)
M6 evaluate_rerun × 3 □□	25 min	$3
M7 per-type + cross-benchmark	15 min	$1
M8 □□ □□ □□ + □□□	25 min	$5
M8 list-prompt □□ □□□	32 min	$7
□□ (□ □□)	~17h GPU + 30h □□	~$49

E.2 □□ □□ □□

Offline cache □ □□□□:

- PPO □□ 10,000 episodes × 32 rollouts × 3 seeds = 960,000 API calls
- □□ □□ $288 (□□□ 1 □□ ~$0.0003)
- □ □□ $33 (cache build) + $0 (PPO) = 85% □□ □□ ($288 → $33)

□□ F. □□·□□□ □□

- □□. GitHub sdw1621/triple-rag-phd (□ □□ □□□), MIT □□□□. □□ □□□ □□ □□ □□.
- PPO □□□□□. cache/ppo_checkpoints/seed_{42,123,999}/final.pt (3 □, □ ~25KB).
- 330K □□ □□. cache/university.sqlite (16.8MB, git-ignored due to size). scripts/build_cache.py □ □□□ □□.
- □□ □□. docs/figures/fig*.png (thesis □ □□□□ □□ □□).
- □□ JSON. results/rerun_*.json (5,000 QA □□ 50 □ □□, □□ □□□ aggregate section).

□□ □□ (□□□□ □□).

```
cd /workspace
python scripts/train_ppo.py --seed 42          # ~16 min on RTX 4090
python scripts/evaluate_rerun.py \\
    --qa data/university/gold_qa_5000.json \\
    --policy ldwa:cache/ppo_checkpoints/seed_42/final.pt \\
    --output results/rerun_ldwa_seed42_list.json \\
    --prompt-style list --workers 10 \\
    --retrieval-cache cache/retrieval_cache.pkl
python scripts/aggregate_list_prompt_results.py
```

□□ □

□□□□ (References)

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